ENVELOPE

Evaluation and validation of connected mobility in real open systems beyond 5GS

Project deliverable D2.2

ENVELOPE platform architecture requirements and specifications

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Legal Disclaimer

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Project executive summary

ENVELOPE aims to advance and open the reference 5G advanced architecture and transform it into a vertical-oriented one. It proposes a novel open and easy-to-use 5G-advanced architecture to enable a tighter integration of the network and the service information domains by

- exposing network capabilities to verticals,
- providing vertical information to the network; and
- enabling verticals to dynamically request and modify key network aspects,

all performed in an open, transparent, and easy-to-use, semi-automated way.

ENVELOPE will build APIs that act as an intermediate abstraction layer that translates the complicated 5GS interfaces and services into easy to consume services accessible by the vertical domain. The project will deliver an experimentation framework that will facilitate vertical services in accessing a series of innovations developed in the project, namely: edge computing with service continuity support (federation/migration), zero-touch management, multi-connectivity, dynamic slicing and predictive QoS.

ENVELOPE will deliver 3 large scale Beyond 5G (B5G) trial sites in Italy, Netherlands and Greece supporting novel vertical services, with advanced exposure capabilities and new functionalities tailored to the services' needs. Although focused on the Connected and Automation Mobility (CAM) vertical, the developments resulting from the use cases (UC) will be reusable by any vertical. The ENVELOPE architecture will serve as an envelope that can cover, accommodate, and support any type of vertical services. The applicability of ENVELOPE will be demonstrated and validated via the project CAM UCs and via several 3rd parties that will have the opportunity to conduct funded research and test their innovative solutions over ENVELOPE.

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Deliverable executive summary

This deliverable, D2.2, presents the architectural framework for the ENVELOPE SNS JU Project. The document outlines the principles and motivations behind the project, emphasizing the importance of openness in 5G and beyond. It provides a comprehensive analysis of the requirements for the ENVELOPE architecture, detailing methodologies and specific needs such as Experimentation as a Service, Management and Orchestration, and Edge Computing. The deliverable explores the architecture enablers and specifications for various components, including B5G Network APIs, ATSSS-like multi-connectivity, and Predictive QoS. Each section is meticulously crafted to ensure that the architecture supports the dynamic and reconfigurable nature of future networks, promoting innovation and efficiency. Key highlights of the deliverable include: i) A thorough background on 5G openness and its evolution. ii) A detailed requirements mapping with use cases to ensure relevance and applicability. iii) Specifications for critical enablers like edge computing and multi-connectivity. iv) An emphasis on predictive quality of service to enhance user experience. The document concludes with a summary of the findings and recommendations, aiming to guide the design phase and identify the appropriate means to address the identified requirements.

List of abbreviations and acronyms

Acronym	Meaning
3GPP	3rd Generation Partnership Project
5GAA	5G Automotive Association
5GS	5G System
5QIs	5G QoS Indicators
API	Application Programming Interface
ATSSS	Access Traffic Steering, Switching, and Splitting
AV	Automated (Driving) Vehicle
B5G	Beyond 5G
CAM	Connected Automated Mobility
CAPIF	Common API Framework
CCAM	Cooperative Connected Automated Mobility
CPM	Cooperative Perception Message
CPU	Central processing unit
DENM	Decentralized Environmental Notification Message
DNN	Data Network Name
DT	Digital Twin
ECU	Electronic Control Unit
EC	European Commission
EAS	Edge Application Server
EES	Edge Enabling Server
eMBB	Enhanced Mobile Broadband
ESU	Electronic Storage Unit
ETSI	European Telecommunications Standards Institute

FR	Functional Requirement
GDPR	General Data Protection Regulation
gNB	gNodeB
GBR	Guaranteed Bit Rate
GNSS	Global Navigation Satellite System
GPU	Graphical Processing Unit
HD	High Definition
HPLMN	Home Public Land Mobile Network
ITS	Intelligent Transportation Systems
K8S	Kubernetes
K3S	Reference abbreviation for a lightweight K8S
LBO	Local break out
LHTI	Local Hazard and Traffic Information
MNO	Mobile Network Operator
NFR	Non-Functional Requirement
OTT	Over-The-Top
OBU	On Board Unit
PCF	Policy Control Function
PDU	Power Distribution Unit
PNI-NPN	Public Network Integrated Non-Public Network
PLMN	Public Land Mobile Network
pQoS	Predictive Quality of Service
QoS	Quality of Service
RAN	Radio Access Network
SDN	Software-Defined Networking
SDO	Standardization Development Organization

SEPP	Security Edge Protection Proxy
SMF	Session Management Function
SSC	Service and Session Continuity
UC	Use Case
UDM	Unified Data Management
UDR	Unified Data Repository
UE	User Equipment
UPF	User Plain Function
VPLMN	Visited Public Land Mobile Network
VRU	Vulnerable Road Users
V2N	Vehicle-to-Network
V2X	Vehicle-to-Everything

1 Introduction

1.1 Objective of the document

The objective of Deliverable D2.2 is to outline the architectural framework for the ENVELOPE SNS JU Project. This document aims to define the principles and motivations behind the project, emphasizing the importance of openness in 5G and beyond. It provides a comprehensive analysis of the requirements for the ENVELOPE architecture, detailing technological features, such as Experimentation as a Service, Management and Orchestration, and Edge Computing that are envisioned within the scope of the project. The deliverable explores the architecture enablers and specifications for various components, interfaces, ensuring that the architecture supports the dynamic and reconfigurable nature of future networks, promoting innovation and efficiency, in a series of corresponding functional capabilities e.g., ATSSS-like multi-connectivity, edge computing federation, Predictive QoS,etc.

1.2 Structure of the document

D2.2 structure ensures a comprehensive and detailed presentation of the ENVELOPE architecture, covering all necessary aspects from principles and requirements to specific enablers and their specifications. More specifically:

1. Introduction: This section outlines the objective, structure, and target audience of the document.

2. ENVELOPE Principles and Motivation: It provides background on 5G openness, defines elementary terms, and explains the ENVELOPE approach.

3. Requirements for the ENVELOPE Architecture: This part details the methodology used, specific requirements for various aspects such as Experimentation as a Service, Management and Orchestration, Edge Computing, and more. It also includes a mapping of requirements with use cases.

4. Architecture Enablers and Specifications for CAM: This section introduces the enablers for the architecture, including Experimentation as a Service, Management and Orchestration, B5G Network APIs, Edge Computing, ATSSS-like multi-connectivity, Predictive QoS, and Roaming. Each enabler is discussed in detail, including state-of-the-art overviews, architectural specifications, and relevant APIs.

5. Conclusions: The document concludes with a summary of the findings and recommendations for future work.

1.3 Target Audience

The target audience for Deliverable D2.2 includes researchers, engineers, and professionals involved in the development and deployment of 5G and beyond 5G (B5G) technologies. This document is also intended for consortium partners, stakeholders, and policymakers who are interested in understanding the architectural framework and requirements of the ENVELOPE 6G-

IA SNS Project. Additionally, it serves as a valuable resource for academic institutions, industry experts, and technical committees who are focused on advancing the state-of-the-art in telecommunications and network architecture. The insights provided in this deliverable aim to guide the design and implementation of innovative solutions that address the dynamic and reconfigurable nature of future networks.

2 ENVELOPE principles and motivation

2.1 BACKGROUND ON 5G OPENNESS

The rapid expansion of connectivity, the surge in traffic data, and the diverse range of business models today necessitate the development of highly flexible infrastructures that maintain consistent capabilities and Quality of Service (QoS). In this context, 5G networks must support a variety of new business solutions with differing performance requirements while also enhancing and optimizing existing services compared to previous mobile network generations. Achieving this level of openness requires coordinated efforts between industry stakeholders and standardization bodies.

The 3rd Generation Partnership Project (3GPP) has approached this vision of openness through the Service Based Architecture (SBA) devised for the 5G Core (5GC) network. This new approach facilitates more efficient resource allocation, service orchestration, lifecycle management, and slicing, thereby increasing the network's flexibility, availability, and reliability. This, in turn, enables the exploitation of new business opportunities. In this section, the main 3GPP specifications of TS 23.501 version 16.6.0 Rel. 16 that inspired ENVELOPE project are detailed, along with key standard APIs that will help realize this vision of openness.

The 3GPP 5G system architecture Rel. 16 introduces a core network model that significantly diverges from traditional architectures [15]. To support network fragmentation and foster more dynamic 5G services, it defines an "open" core where all core network functions are virtualized. This approach eliminates resource inefficiencies and performance degradation associated with virtual machines and hypervisors, thereby enhancing the network's flexibility, speed, and automation. A key enabler of this openness is network programmability through standard APIs, allowing higher-level service orchestrators to manage configurations for various services and slices. This initiative creates a new and dynamic ecosystem in mobile networks from both technological and marketing perspectives. Authorized external third parties, such as industries, platform developers, and designers, can use these standard APIs to build network-aware (5G-enabled) applications. These applications establish bi-directional communication with the 5GC, retrieving network statistics and triggering specific policies and commands to the network.

This exposure capability is realized through the Service Based Architecture (SBA) adopted by the 5GC network. The 5GC control plane network functions (NFs) communicate via API calls that define the related Service Based Interfaces (SBIs). In this context, the Network Repository Function (NRF) allows other NFs to register their services, which can then be discovered by other NFs. This enables a versatile implementation where each NF can access resources from other approved NFs. Key benefits of API exposure in 5G networks include:

- **Enhanced Service Innovation**: By exposing network capabilities through APIs, 5G networks enable third-party developers to create innovative applications that leverage the advanced features of the network. This can lead to the development of new services in areas such as IoT, smart cities, autonomous vehicles, and more.
- **Improved Network Management**: APIs provide a standardized way to monitor and manage network functions, enabling more efficient and automated network operations. This

can lead to improved network performance, reduced downtime, and enhanced user experience.

- **Greater Flexibility and Customization**: API exposure allows network operators to offer customized services tailored to the specific needs of different verticals. This flexibility can help meet the diverse requirements of industries such as healthcare, manufacturing, and transportation.

Additionally, the Network Exposure Function (NEF) provides a set of northbound APIs for exposing network data and receiving management commands. Specifically, NEF offers adaptors that connect the southbound interfaces with the SBA to an exposure layer with northbound interfaces available to third-party developers. This approach, illustrated in Figure 1, facilitates the secure disclosure of network resources to third parties, such as network slicing, edge computing, and machine learning utilizing the 5G system. This is fully compliant with the innovative paradigms underpinning a wide range of services.

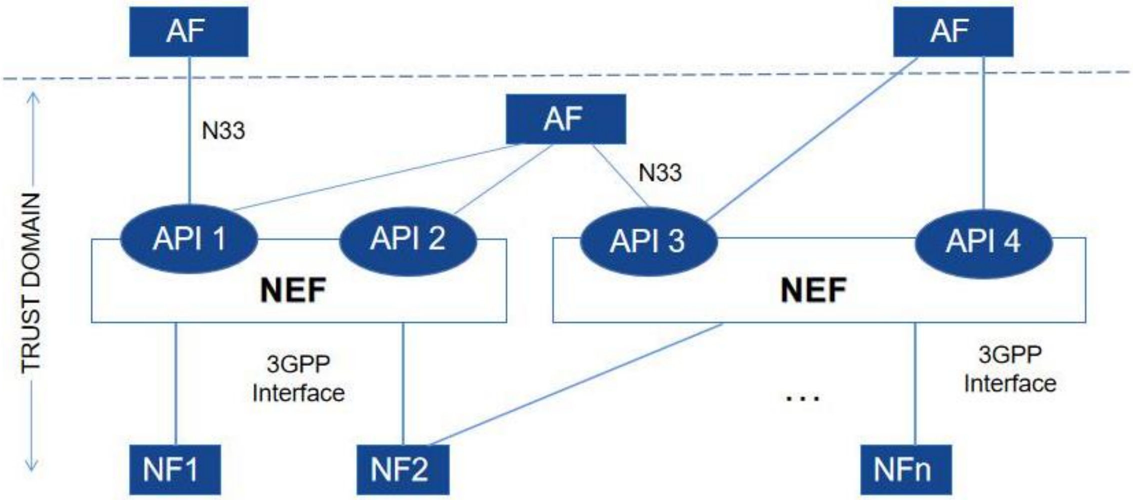

Figure 1 – 5G Core network exposure

Respectively, the Network Data Analytics Function (NWDAF) is a key component in the 5G core network architecture, specified by the 3rd Generation Partnership Project (3GPP) [26]. NWDAF plays a crucial role in enabling advanced data analytics within the 5G network, providing insights into network performance, user behaviour, and other relevant data. Some key functionalities of NWDAF include Data Collection, where NWDAF collects data from various sources within the 5G network, including user equipment (UE), gNB (Next-Generation NodeB), and other network elements. Respectively, NWDAF employs advanced analytics algorithms to process the collected data. Machine learning and artificial intelligence techniques may be used to identify patterns, anomalies, and trends in the data. NWDAF focuses on generating Key Performance Indicators that reflect the health and efficiency of the 5G network. KPIs may cover aspects such as network latency, throughput, reliability, and resource utilization.

This openness to third parties enables the programmability and adaptability of 5G connectivity services, creating a new ecosystem where third-party developments bridge 5G exposed capabilities and service requirements/potentials from vertical industries. This feature of the 5G core is the foundation upon which ENVELOPE builds its innovation.

3GPP Rel.17 brought significant enhancements to the 5G system, focusing on improving performance and supporting new use cases and verticals. Respectively, Rel. 18 marks the start of 5G-Advanced by introducing further intelligence into wireless networks by implementing machine-learning-based techniques at different levels of the network. Release 18 also emphasized the importance of network automation and sustainable networks. Finally, Rel. 19 is the second release for 5G-Advanced and continues to build on the advancements of its predecessors. By incorporating developments from Releases 17, 18, and 19, the ENVELOPE project ensures that its architecture remains at the cutting edge of 5G technology. These enhancements provide the necessary foundation for supporting advanced vertical applications, including CAM, and pave the way for the next generation of mobile communication systems.

2.2 ENVELOPE Elementary Terms

In the following we introduce a series of term definitions underlying the definition of the ENVELOPE requirements and architecture.

ENVELOPE APIs

The ENVELOPE APIs are a set of APIs that are implemented within the context of the ENVELOPE project, and they are made available through the ENVELOPE Platform. The ENVELOPE APIs are accessed by a Vertical Service to interact with the Beyond 5G System. They enable a Vertical Service to retrieve network information and influence network configuration.

ENVELOPE Enabler

An ENVELOPE Enabler is an add-on to the 5G System provided by the ENVELOPE Platform. The ENVELOPE Enabler makes available a major advance in the Beyond 5G System implementing a specific feature such as Predictive QoS, dynamic slicing, AF traffic influence, ATSSS-like multi-connectivity. The ENVELOPE Enabler allows an experimenter to test advanced Vertical Services requiring 5G features not commonly available in 5G network testbeds.

ENVELOPE Platform

The ENVELOPE platform is a Beyond 5G System experimentation as a service platform with support of simplified and extended interaction with the specific CAM vertical. It encompasses the ENVELOPE APIs, the ENVELOPE Enablers and the Experimentation as a Service framework.

ENVELOPE Facility

The ENVELOPE Facility is a platform for Beyond 5G System experimentation as a service with support of simplified and extended interaction with the specific CAM vertical. It encompasses the ENVELOPE APIs, the ENVELOPE Enablers, the Experimentation as a Service framework and the underlying infrastructure.

Experimentation

The experimentation corresponds to the actions that are needed to define, upload, launch, and evaluate an experiment in an ENVELOPE trial site. The experiment consists of the deployment of a Vertical Service that exploits advanced 5G features enabled by the ENVELOPE platform.

Experimenter

The experimenter is the entity that deploys a Vertical Service in an ENVELOPE trial site for experimenting with some advanced Beyond 5G System's features that are enabled by the ENVELOPE platform. The experimenter is in charge of defining the experiment, launching the experiment, and analyzing the outcomes.

Infrastructure layer

The infrastructure layer is the set of hardware resources that an experimenter can exploit to deploy application functions and network functions. The infrastructure layer comprises far-edge devices, edge and cloud servers.

Trial site

A trial site is the whole set of hardware and software resources that guarantee communication and computing capabilities to an experimenter for running an experiment. The trial site implements the ENVELOPE Platform providing access to ENVELOPE APIs and integrating ENVELOPE Enablers.

Vertical Service

A Vertical Service is composed of interconnected application functions and network functions that implement the application-level functionalities to showcase a specific use case. The Vertical Service interacts with the Beyond 5G System and with the ENVELOPE Enablers through the ENVELOPE APIs. The application functions and the network functions, that constitute a Vertical Service, are distributed on the infrastructure layer ranging from the far edge to the edge and cloud segments.

Predictive QoS

Predictive Quality of Service (QoS) is an advanced feature based on which estimations about the future network or service level performance are generated through means of monitoring data analytics. The estimations are used to trigger notifications towards network or application/service level consuming entities with the purpose of enabling proactive network or application/service level reconfigurations.

AF Traffic Influence

AF (Application Function) Traffic Influence is a mechanism that allows applications to influence the routing and handling of their traffic within the network. This feature enables applications to request specific QoS parameters, prioritize certain types of traffic, and optimize their performance based on the network conditions.

ATSSS Multi-Connectivity

ATSSS (Access Traffic Steering, Switching, and Splitting) Multi-Connectivity is a feature that allows devices to simultaneously connect to multiple network interfaces, such as Wi-Fi and cellular networks. This enhances the reliability and performance of the connection by dynamically steering, switching, and splitting traffic across the available interfaces.

2.3 The ENVELOPE Approach

The ENVELOPE project stands at the forefront of the next evolutionary steps in cellular communications, specifically 5G-Advanced and 6G systems. These evolutionary advancements aim to expand the set of supported vertical use cases and provide enhanced capabilities beyond mere connectivity. The integration of existing and future vertical applications with the network, coupled with the complete automation of network and service management, marks a paradigm shift from performance-driven to value-driven networks.

5G-Advanced and 6G are not just incremental upgrades but revolutionary steps that promise to transform various sectors, including Connected and Automated Mobility (CAM). The global connected car market, projected to reach USD 56.3 billion by 2026, underscores the significant economic and societal impact of connected mobility. Factors such as the increasing trend of in-vehicle connectivity solutions, the development of smart city infrastructure, 5G infrastructure, and intelligent transportation systems (ITS) are driving this growth. Developments in autonomous vehicles, growing electric vehicle sales, and the popularity of integrated/smartphone-connected solutions are expected to further enhance car connectivity. Mobile communication systems like 5G and 6G will play a central role in the future transport ecosystem, requiring a reliable and safe guidance infrastructure that combines all available technologies: sensors, converged AI on devices, at the edge and in the cloud, and high-quality communications between all moving and fixed elements.

The 5G System (5GS) was initially designed as a modular architecture to support any vertical running on top in a vertical-agnostic manner. However, it soon became evident that certain verticals have specific requirements that need to be addressed. For instance, the automotive sector's challenge of handling "big data" has not been fully addressed within the current 3GPP framework. This gap poses a risk that future network deployments and business models may fail to support the emerging needs of Connected and Automated Vehicles (CAVs). Subsequent releases of 5GS (up to Rel-18) have focused on enabling the necessary functionalities for a diverse set of verticals, such as Location services, Edge Computing (EDC), and Network Slicing (NS).

Despite significant progress, the configuration of the network and end-devices (UEs) to support a vertical remains a time-consuming manual process. This process requires tight coordination at technical and business levels across the verticals, the UE vendor, the network operator, and even the end-user. This complexity hinders the greater adoption of 5GS and the uptake of novel CAM vertical use cases, as well as the digitalization and modernization of existing ones.

Given the limited market penetration of 5GS in vertical domains, further work is needed to identify and resolve the pain points of 5GS towards its evolution to 5G-Advanced and beyond. This is critical for services like CAM and ITS that require tighter integration and interaction with the underlying network. Key requirements include high mobility, cross-border and cross-domain operation, demanding and dynamically changing QoS/QoE requirements, URLLC requirements, multi-connectivity, dynamic access selection, selective traffic redundancy, and dynamic network slice selection and adjustment.

The main objective of the ENVELOPE project is to advance and open up the reference 5G-Advanced architecture, transforming it into a vertical-oriented one with the necessary interfaces tailored to CAM vertical use cases. The project aims to:

1. **Expose Network Capabilities to Verticals**: Provide verticals with better visibility of the underlying network/communication state.
2. **Provide Vertical-Information to the Network**: Enable the network to receive and utilize information from verticals to optimize performance.

3. **Enable Verticals to Dynamically Request and Modify Network Aspects**: Allow verticals to influence network configuration in an open, transparent, and semi-automated way.

The envisioned 5G evolution comes with new challenges, including unknown KPIs/KVIs, the need for testing and verification of vertical services in evolved infrastructures, and the necessity of openness and a vertical abstraction layer. This layer will promote transparent interaction between the network and the vertical. A necessary prior step is to identify the business and technical limitations of existing systems for critical applications like CAM and ITS services through validation in large-scale trials and pilots before enabling services in the real world.

ENVELOPE will build on the success of previous CAM-related and SNS projects, aiming to deliver three large-scale Beyond 5G (B5G) trial site infrastructure deployments in **Italy**, the **Netherlands**, and **Greece** for CAM vertical services and beyond. These deployments will be enhanced with functionalities tailored to CAM services and advanced exposure capabilities. Although ENVELOPE focuses on the CAM and intelligent terrestrial transportation vertical, the resulting developments and lessons learnt will be reusable by any vertical. **The proposed ENVELOPE architecture in this deliverable will serve as an envelope that can cover, accommodate, and support any type of vertical service**. The applicability of ENVELOPE capabilities in different vertical domains will be demonstrated through the project's CAM-oriented use cases and at least nine open call projects to be funded during the project.

The ENVELOPE project is a critical step towards realizing the full potential of 5G-Advanced and 6G technologies. By addressing the specific needs of verticals like CAM and ITS, and by creating a flexible, modular, and vertical-oriented architecture, ENVELOPE aims to drive the adoption and evolution of 5G technologies. Through its innovative approach and collaborative efforts, ENVELOPE is set to pave the way for the next generation of mobile communication systems and their integration into various vertical domains.

3 Requirements for the ENVELOPE architecture

3.1 Methodology

For any project, it's crucial to define the objectives from the outset, essentially setting the project's scope. To achieve this, a detailed analysis of the use case scenarios to be addressed in the project has been conducted in D2.1. Following this, the features and functionalities of the relevant use case scenarios have been identified and formalized as requirements of the ENVELOPE architecture in this section.

While various definitions exist of what is a requirement, ENVELOPE agreed to use the ISO/IEC 2007 definition: "*A requirement is Statement that identifies a product (includes product, service, or enterprise) or process operational, functional, or design characteristic or constraint, which is unambiguous, testable or measurable, and necessary for product or process acceptability.*"

Once the definition was clarified, the process of documenting a technical requirement in ENVELOPE adheres to principles outlined in the Volere requirements specification methodology [28], which is believed to result in a more thorough and precise record. The following guiding principles were also agreed by partners of the project:

Independently of the type of requirement, those can be classified in two categories:

Functional requirements: are the fundamental subject matter of the system and are expressed / realized by decision-making logic, and algorithms.

Non-functional requirements: are the behavioural properties that the specified functions must have, such as performance, usability, etc. Non-functional requirements can be assigned to a specific measurement.

Then the requirements were grouped in different categories, based on the building block of ENVELOPE system that each one of them refers to. More specifically, the following categories were defined:

- **Portal:** Referring to the ENVELOPE experimentation portal.
- **Application repository:** Referring to the repository of the applications.
- **Experimentation:** Referring to the experimentation lifecycle.
- **Monitoring platform:** Referring to the monitoring process.
- **Trial site capabilities repository:** Referring to the trial site capabilities.
- **Management and Orchestration:** Referring to the MANO aspects.
- **B5G System:** Referring to the beyond 5G system capabilities.
- **Network API:** Referring to the APIs used for the interaction with 5GS NFs.
- **Exposure Framework**: Referring to framework used for exposing Network APIs
- **EDGE System:** Referring to the edge computing system .
- **Multi-Connectivity:** Referring to the multi-connectivity capabilities.
- **PQoS:** Referring to the Predictive QoS aspects
- **ENVELOPE API:** Referring to the APIs exposed by the ENVELOPE architecture towards the vertical services.
- **Roaming:** Referring to the inter-PLMN mobility and corresponding roaming functionalities.

In addition, a common requirement definition criterion adopted in ENVELOPE was the prioritization. Project partners decided to follow the requirements prioritization classifying them as:

- **REQUIRED**, when an ENVELOPE platform MUST implement the specific requirement.
- **RECOMMENDED**, when the trial sites MAY implement the specific requirement. Remaining focused on tangible targets, the project ensures that at least one trial site addresses the specific requirement.

Where the use of MUST and MAY is the above definitions comply with the IETF RFC 2119 methodology, meaning that:

- **MUST** This word, or the terms "REQUIRED" or "SHALL", mean that the definition is an absolute requirement of the specification.
- **MAY** This word, or the adjective "OPTIONAL", mean that an item is truly optional.

The project adopted an agile methodology for requirements analysis, adhering to industry best practices. To ensure the involvement of the right stakeholders and full engagement from all consortium partners, requirements were reviewed during weekly calls from the project's inception. Brainstorming sessions were supported by the '*Record Information*' process, utilizing the project's cloud repository and a collaboratively edited Excel file with a structured, type-checked format. Requirements were clearly defined as shown in the following reference Table, and experts in each technical area led the '*Rinse and Repea*t' process to ensure thoroughness and accuracy. The aim was not to create an exhaustive list of technical details, but to identify key requirements to guide the design phase and determine appropriate solutions. Below we provide the template table used to capture the requirements definitions.

ID: REQ-*Relevant WP3 Task Description*_FN_ID	Name:	
Type: Functional Requirements | Non-Functional Requirements	Category: Portal | Application repository | Experimentation | Monitoring platform | Trial site capabilities repository | Management and Orchestration | B5G System | Network API | Exposure Framework | EDGE System | Multi-Connectivity | PqoS | Network API | Roaming	Priority: REQUIRED RECOMMENDED
Description A brief description of the requirement		

3.2 ENVELOPE requirements

3.2.1 Experimentation as a Service

ID: REQ-F-Task3.1-Eaas-1	Name: ENVELOPE Portal – authentication	
Type: FR	Category: Portal	Priority: Required
Description: The ENVELOPE portal shall authenticate the experimenter		

ID: REQ-F-Task3.1-Eaas-2	Name: ENVELOPE Portal – experimentation lifecycle interface	
Type: FR	Category: Portal	Priority: Required
Description: The ENVELOPE portal shall provide the graphical interface for allowing the experimenter to interact with the Experiment lifecycle manager		

ID: REQ-F-Task3.1-Eaas-3	Name: ENVELOPE Portal – monitoring interface	
Type: FR	Category: Portal	Priority: Required
Description: The ENVELOPE portal shall provide the graphical interface for allowing the experimenter to interact with the monitoring tools		

ID: REQ-F-Task3.1-Eaas-4	Name: ENVELOPE Portal – application repository interface	
Type: FR	Category: Portal	Priority: Required
Description: The ENVELOPE portal shall provide the graphical interface for allowing the experimenter to interact with the Application repository (see detailed interaction requirements below)		

ID: REQ-F-Task3.1-Eaas-5	Name: ENVELOPE Portal – trial site capabilities repository interface	
Type: FR	Category: Portal	Priority: Required
Description: The ENVELOPE portal shall provide the graphical interfaces for allowing the experimenter to interact with the trial site capabilities repository		

<table>
<tr><td>ID: REQ-F-Task3.1-Eaas-6</td><td colspan="2">Name: Application repository – loading application</td></tr>
<tr><td>Type: FR</td><td>Category: Application repository</td><td>Priority: Required</td></tr>
<tr><td colspan="3">Description: The experimenter shall have the possibility to load applications in the Application repository</td></tr>
</table>

<table>
<tr><td>ID: REQ-F-Task3.1-Eaas-7</td><td colspan="2">Name: Application repository – application availability</td></tr>
<tr><td>Type: FR</td><td>Category: Application repository</td><td>Priority: Required</td></tr>
<tr><td colspan="3">Description: The experimenter shall have the possibility to check which applications have been loaded</td></tr>
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<table>
<tr><td>ID: REQ-F-Task3.1-Eaas-8</td><td colspan="2">Name: Application repository – updating application</td></tr>
<tr><td>Type: FR</td><td>Category: Application repository</td><td>Priority: Required</td></tr>
<tr><td colspan="3">Description: The experimenter shall have the possibility to update applications in the Application repository</td></tr>
</table>

<table>
<tr><td>ID: REQ-F-Task3.1-Eaas-9</td><td colspan="2">Name: Application repository – deleting application</td></tr>
<tr><td>Type: FR</td><td>Category: Application repository</td><td>Priority: Required</td></tr>
<tr><td colspan="3">Description: The experimenter shall have the possibility to delete the applications that have been loaded</td></tr>
</table>

<table>
<tr><td>ID: REQ-F-Task3.1-Eaas-10</td><td colspan="2">Name: Application repository – public/private application</td></tr>
<tr><td>Type: FR</td><td>Category: Application repository</td><td>Priority: Required</td></tr>
<tr><td colspan="3">Description: The experimenter shall have visible only its own applications if these have not set as public during the loading phase</td></tr>
</table>

ID: REQ-F-Task3.1-Eaas-11	Name: Experimentation descriptor – application chaining	
Type: FR	Category: Experimentation	Priority: Required
Description: The experimentation descriptor shall allow to indicate the applications chain making up the experimentation		

ID: REQ-F-Task3.1-Eaas-12	Name: Experimentation descriptor – computing resources	
Type: FR	Category: Experimentation	Priority: Required
Description: The experimentation descriptor shall allow to indicate the amount of computing resources (i.e., storage, RAM, CPU) needed by each application in the experimentation		

ID: REQ-F-Task3.1-Eaas-13	Name: Experimentation descriptor – communication resource	
Type: FR	Category: Experimentation	Priority: Required
Description: The experimentation descriptor shall allow to indicate the amount of network resources (e.g., type of network slice or 5QI) needed by each application in the experimentation		

ID: REQ-F-Task3.1-Eaas-14	Name: Experiment lifecycle manager – experimentation APIs	
Type: FR	Category: Experimentation	Priority: Required
Description: The Experiment lifecycle manager shall make available a set of experimentation APIs for managing the lifecycle of the experimentation		

ID: REQ-F-Task3.1-Eaas-15	Name: Experiment lifecycle manager – descriptor repository	
Type: FR	Category: Experimentation	Priority: Required
Description: The Experiment lifecycle manager shall allow to store the descriptor of a given experimentation for later (re)use by the experimenter		

ID: REQ-F-Task3.1-Eaas-16	Name: Experiment lifecycle manager – computing nodes	
Type: FR	Category: Experimentation	Priority: Required
Description: The Experiment lifecycle manager shall provide the computing nodes (i.e., far edge nodes, edge and cloud servers) on which it is possible to deploy applications		

ID: REQ-F-Task3.1-Eaas-17	Name: Experiment lifecycle manager – computing resources	
Type: FR	Category: Experimentation	Priority: Required
Description: The Experiment lifecycle manager shall request to the compute orchestrator the computing resources for each application of the experimentation as requested by the experimenter in the descriptor of the experimentation		

ID: REQ-F-Task3.1-Eaas-18	Name: Experiment lifecycle manager – network resources	
Type: FR	Category: Experimentation	Priority: Required
Description: The Experiment lifecycle manager shall request to the network orchestrator the networking resources for each application of the experimentation as requested by the experimenter in the descriptor of the experimentation		

ID: REQ-F-Task3.1-Eaas-19	Name: Experimentation APIs – load experimentation	
Type: FR	Category: Experimentation	Priority: Required
Description: The Experimentation APIs shall include an API for loading the experimentation. In the loading phase, the experiment lifecycle manager checks the feasibility of the experimentation, and reserves the needed resources.		

ID: REQ-F-Task3.1-Eaas-20	Name: Experimentation APIs – start experimentation	
Type: FR	Category: Experimentation	Priority: Required

Description: The Experimentation APIs shall include an API for starting the experimentation. At the start of the experimentation, the applications are deployed (i.e., loaded and executed) on the computing nodes.

ID: REQ-F-Task3.1-Eaas-21	Name: Experimentation APIs – stop experimentation	
Type: FR	Category: Experimentation	Priority: Required
Description: The Experimentation APIs shall include an API for stopping the experimentation. At the end of the experimentation, applications are stopped and resources are freed.		

ID: REQ-F-Task3.1-Eaas-22	Name: Monitoring platform – metrics management	
Type: FR	Category: Monitoring platform	Priority: Required
Description: The Monitoring platform shall collect, store and make available the metrics that an experiment wants as outcomes of the experimentation.		

ID: REQ-F-Task3.1-Eaas-23	Name: Trial site capabilities repository	
Type: FR	Category: Trial site capabilities repository	Priority: Required
Description: The Trial site capabilities repository shall store and make available the capabilities of the trial site.		

3.2.2 Intent-Driven Management and Orchestration

ID: REQ-F-Task3.5-ZTM-1	Name: Intent translation	
Type: FR	Category: Management and Orchestration	Priority: Recommended
Description: Transform the intent into optimization objectives.		

ID: REQ-F-Task3.5-ZTM-2	Name: Orchestration and execution of Decisions	
Type: FR	Category: Management and Orchestration	Priority: Recommended
Description: Implement a solution for orchestration and execution of Decisions related to specific CAM application parameters (e.g. Anomaly Detection sensitivity) via the automatic generation of control loops with embedded intelligence.		

ID: REQ-F-Task3.5-ZTM-3	Name: Intent APIs	
Type: FR	Category: Management and Orchestration	Priority: Recommended
Description: Establish intent APIs that link the Portal with the Decision Engine (DE) responsible for handling the intents and controlling specific CAM application parameters based on the latter as well as telemetry data.		

ID: REQ-F-Task3.5-ZTM-4	Name: Resource Orchestration mechanisms	
Type: FR	Category: Management and Orchestration	Priority: Recommended
Description: Implement mechanisms to handle the lifecycle of the computing resources, to be used as input for the decision making system responsible for controlling specific CAM application parameters.		

ID: REQ-F-Task3.5-ZTM-5	Name: Intelligent resource allocation system

Type: FR	Category: Management and Orchestration	Priority: Recommended
Description: Intelligent system for selecting the allocation strategy from the requirements specified in the service instantiation intent request.		

ID: REQ-F-Task3.5-ZTM-6	Name: Dynamic Service (re-)configuration through Closed Loop Systems	
Type: FR	Category: Management and Orchestration	Priority: Recommended
Description: The Orchestration system should be able to instantiate Closed Loop functions for the automatic (re-)configuration of the network and the services.		

ID: REQ-F-Task3.5-ZTM-7	Name: Context-aware decision making system	
Type: FR	Category: Management and Orchestration	Priority: Recommended
Description: The Orchestrator should be able to make decisions based on monitoring metrics gathered from the system (i.e., service-specific metrics, computing resources monitoring, network metrics, position of far-edge devices, etc.).		

3.2.3 Open and dynamic reconfigurable B5G System (B5GS)

ID: REQ-F-Task3.2-B5GS-1	Name: APIs exposure to untrusted applications – N33	
Type: FR	Category: B5GS	Priority: Required
Description: 5GS APIs must be exposed to untrusted applications through NEF via N33 interface as defined in [3GPP TS 23.501]		

ID: REQ-F-Task3.2-B5GS-2	Name: APIs exposure to trusted applications – N5	
Type: FR	Category: B5GS	Priority: Required
Description: 5GS APIs must be exposed to trusted applications through PCF via N5 interface as defined in [3GPP TS 23.501]		

ID: REQ-F-Task3.2-B5GS-3	Name: NEF authentication

Type: FR	Category: B5GS	Priority: Recommended
Description: Mutual authentication between NEF and Visitor Serving Access and Mobility Function (VS AF) must be supported [3GPP TS 33501]		

ID: REQ-F-Task3.2-B5GS-4	Name: NEF integrity, relay, confidentiality protection	
Type: FR	Category: B5GS	Priority: Recommended
Description: Integrity, relay and confidentiality protection must be supported for the communication between NEF and VS AF [3GPP TS 33501]		

ID: REQ-F-Task3.2-B5GS-5	Name: NEF authorization	
Type: FR	Category: B5GS	Priority: Recommended
Description: NEF must determine whether the VS AF is authorized to interact with the relevant APIs [3GPP TS 33501]		

ID: REQ-F-Task3.2-B5GS-6	Name: E2E Latency measurement API	
Type: FR	Category: B5GS	Priority: Recommended
Description: Design and implement open APIs exposing the measurement of E2E data transmission quality (e.g., E2E latency) between the client (UE) and the server in the DN		

ID: REQ-F-Task3.2-B5GS-7	Name: QoS 30prioritization30 with GBR support	
Type: FR	Category: B5GS	Priority: Recommended
Description: 5GS should support QoS differentiation with GBR support end-to-end		

ID: REQ-F-Task3.2-B5GS-8	Name: Dynamic UE configuration	
Type: FR	Category: B5GS	Priority: Recommended
Description: 5GS should support for dynamic UE configuration updates via URSPs		

ID: REQ-F-Task3.2-B5GS-9	Name: Containarised and Virtualised Network Functions implementations	
Type: FR	Category: B5GS	Priority: Recommended

Description: To prioritise the dynamic scale-in and out of B5GS the appropriate prioritization capabilities must be considered, such as the containerization and virtualization of the NFs.

Network APIs

ID: REQ-*F-Task3.2-B5GS -10*	Name: Quality on Demand (QoD)	
Type: FR	Category: Network API	Priority: Required
Description: Support for dynamic (on-demand) QoS reconfiguration for connected mobile devices (UEs) is required		

ID: REQ-*F-Task3.2-B5GS -11*	Name: Device Location	
Type: FR	Category: Network API	Priority: Required
Description: Subscribing to device location updates should be supported		

ID: REQ-*F-Task3.2-B5GS -12*	Name: Discovery of optimal MEC service instance	
Type: FR	Category: Network API	Priority: Required
Description: Discovery of optimal MEC service instance (e.g., given location, services supported, QoS requirements) should be supported		

ID: REQ-*F-Task3.2-B5GS -13*	Name: Dynamic UE slice configuration	
Type: FR	Category: Network API	Priority: Recommended
Description: The system should support network slice information delivery to UEs for dynamic UE slice configuration updates [NOTE: A SEAL approach is promoted i.e., 3GPP TS 23.435]		

ID: REQ-*F-Task3.2-B5GS -14*	Name: Traffic Influence	
Type: FR	Category: Network API	Priority: Required
Description: Dynamic re-configuration of the optimal routing from the user device to the optimal application instance in a specific geographical location, install in an EdgeCloud Zone should be supported		

ID: REQ-F-Task3.2-B5GS-15	Name: AF Traffic Influence API	
Type: FR	Category: Network API	Priority: Recommended
Description: NEF must expose AF Traffic Influence API through N33 interface to untrusted VS AF [3GPP TS 29.522]		

ID: REQ-F-Task3.2-B5GS-16	Name : Dynamic UE QoS PDU session configuration	
Type: FR	Category: Network API	Priority: Recommended
Description: NEF should expose AsSessionWithQoS API through N33 interface to untrusted VS AF [3GPP TS 29.522]		

ID: REQ-F-Task3.2-B5GS-17	Name: UE location/mobility information	
Type: FR	Category: Network API	Priority: Recommended
Description: NEF should expose UE location/mobility information via EventExposure or MonitoringEvent APIs through N33 interface to untrusted VS AF [3GPP TS 29.591, 3GPP 29.522]		

ID: REQ-F-Task3.2-B5GS-18	Name: Provisioning requests	
Type: FR	Category: Network API	Priority: Recommended
Description: B5GS should be able to process the provisioning and slice configuration requests (QoS, DNNs, edge location) generated by an AF		

ID: REQ-F-Task3.2-B5GS-19	Name: Provisioning APIs	
Type: FR	Category: Network API	Priority: Recommended
Description: B5GS shall support Unified Data Management and Unified Data Repository (UDM/UDR) provisioning APIs to be able to modify the SUPIs profiles including QoS parameters		

ID: REQ-F-Task3.2-B5GS-20	Name: Service management APIs

Type: FR	Category: Network API	Priority: Recommended
Description: B5GS shall support the service management APIs to switch on/off core network services on-demand		

ID: REQ-F-Task3.2-B5GS-21	Name : Dynamic UE QoS PDU session configuration	
Type: FR	Category: Network API	Priority: Recommended
Description: B5GS should expose PostAppSessions APIs through N5 interface to trusted VS AF [TS 29.514]		

ID: REQ-F-Task3.2-B5GS-22	Name: Network metrics exposure	
Type: FR	Category: Network API	Priority: Recommended
Description: B5GS shall expose network-side monitoring information about packet level QoS metrics		

ID: REQ-NF-Task3.2-B5GS-23	Name: Dynamic Network Slicing API	
Type: FR	Category: Network API	Priority: Recommended
Description: Each site should support dynamic network slicing, allowing the creation of customized virtual networks to meet the specific requirements of the experimenter		

ID: REQ-NF-Task3.2-B5GS-24	Name: NEF authentication	
Type: NF	Category: Network API	Priority: Required
Description: Mutual authentication based on client and server certificates shall be performed between the NEF and VS AF using TLS [3GPP TS 33501]		

ID: REQ-NF-Task3.2-B5GS-25	Name: NEF integrity, relay, confidentiality protection

Type: NF	Category: Network API	Priority: Required
Description: TLS shall be used by the NEF to provide integrity, relay and confidentiality protection [3GPP TS 33501]		

ID: REQ-NF-Task3.2-B5GS-26	Name: NEF authorisation	
Type: NF	Category: Network API	Priority: Required
Description: NEF shall support Oauth-based authorization mechanism [3GPP TS 33501]		

ID: REQ-NF-Task3.2-B5GS-27	Name: 3GPP SUPI	
Type: NF	Category: Network API	Priority: Required
Description: SUPI shall not be sent outside the 3GPP operator domain [3GPP TS 33501]		

ID: REQ-NF-Task3.2-B5GS-28	Name: Management API security	
Type: NF	Category: Network API	Priority: Required
Description: Management APIs exposed by B5GS should rely on TLS		

Exposure Framework

ID: REQ-F-Task3.2-B5GS-29	Name: Discover_Service_API	
Type: FR	Category: Exposure Framework	Priority: Recommended
Description: The CAPIF discover service APIs, as defined in 3GPP TS 23.222		

ID: REQ-F-Task3.2-B5GS-30	Name: Publish_Service_API	
Type: FR	Category: Exposure Framework	Priority: Recommended
Description: The CAPIF publish service APIs, as defined in 3GPP TS 23.222		

ID: REQ-F-Task3.2-B5GS-31	Name: API_Invoker_Management_API	
Type: FR	Category: Exposure Framework	Priority: Recommended
Description: The CAPIF API invoker management APIs, as defined in 3GPP TS 23.222		

ID: REQ-F-Task3.2-B5GS-32	Name: API_Provider_Management_API	
Type: FR	Category: Exposure Framework	Priority: Recommended
Description: The CAPIF API provider management APIs, as defined in 3GPP TS 23.222		

ID: REQ-F-Task3.2-B5GS-33	Name: Logging_API_Invocation_API	
Type: FR	Category: Exposure Framework	Priority: Recommended
Description: The Logging API invocations APIs, as defined in 3GPP TS 23.222		

ID: REQ-F-Task3.2-B5GS-34	Name: Events_API	
Type: FR	Category: Exposure Framework	Priority: Recommended
Description: The CAPIF events APIs, as defined in 3GPP TS 23.222		

ID: REQ-F-Task3.2-B5GS-35	Name: Security_API	
Type: FR	Category: Exposure Framework	Priority: Recommended
Description: The CAPIF security APIs, as defined in 3GPP TS 23.222		

ID: REQ-F-Task3.2-B5GS-36	Name: Auditing_API	
Type: FR	Category: Exposure Framework	Priority: Recommended
Description: The Auditing API, as defined in 3GPP TS 23.222		

3.2.4 Edge Computing

ID: REQ-F-Task3.4-MEC-1	Name: MEC service discovery	
Type: FR	Category: EDGE System	Priority: Required
Description: Discovery of services available in the MEC platform should be supported (e.g., ETSI MEC)		

ID: REQ-F-Task3.4- MEC-2	Name: MEC federation	
Type: FR	Category: EDGE System	Priority: Required
Description: MEC federation across different service/edge domains should be supported (e.g., ETSI MEC) [29]		

ID: REQ-F-Task3.4-MEC-3	Name: MEC handover	
Type: FR	Category: EDGE System	Priority: Required
Description: Handover of applications among edge servers should be supported (e.g., ETSI MEC)		

ID: REQ-F-Task3.4-MEC-4	Name: MEC service registration	
Type: FR	Category: EDGE System	Priority: Recommended
Description: Dynamic registration of a service in the MEC service registry.		

ID: REQ-F-Task3.4-MEC-5	Name: UE traffic (re-)routing to optimal edge UPF	
Type: FR	Category: EDGE System	Priority: Recommended
Description: UE traffic re-routing to optimal edge UPF given service requirements. For example, by changing UE's policy to new slice (S-NSSAI) and/or DNN.		

ID: REQ-F-Task3.4-MEC-6	Name: MEC host discovery	
Type: FR	Category: EDGE System	Priority: Recommended
Description: Discovery of optimal edge host available given specific service requirements (e.g., GPU available, processing, memory)		

3.2.5 Roaming

ID: REQ-F-Task3.4-MEC-7	Name: Time synchronisation	
Type: FR	Category: Roaming	Priority: Recommended

Description: Cross-domain time 3737synchronisation using the same clock reference is necessary to support the application operation

ID: REQ-F-Task3.4-MEC-8	Name: During network hand-overs the active sessions must be maintained	
Type: FR	Category: Roaming	Priority: Recommended
Description: The implementation of seamless inter-PLMN handover, including core-initiated handover to ensure session migration instead of re-establishment.		

ID: REQ-F-Task3.4-MEC-9	Name: At the radio-level TDD operations of adjacent MNOs must be synchronised	
Type: FR	Category: Roaming	Priority: Recommended
Description: For networks using 5G midband with TDD operations, neighboring operators must follow GSMA and ECC/CEPT recommendations		

ID: REQ-F-Task3.4-MEC-10	Name: Network Reselection Improvements	
Type: FR	Category: Roaming	Priority: Recommended
Description: To assist seamless handover neighboring relations in the edge gNBs must be appropriately defined		

3.2.6 Predictive QoS

ID: REQ-F-Task3.3-INTER-1	Name: Support QoS Sustainability and DN performance predictive analytics notifications	
Type: FR	Category: PQoS	Priority: Recommended
Description: The 5GS shall support QoS Sustainability and DN performance predictive analytics notifications towards subscribed entities at the service layer (e.g., AF), through e.g., NEF or NWDAF.		

ID: REQ-F-Task3.3-INTER-2	Name: 5GS support for Over-The-Top (OTT) predictive performance analytics	
Type: FR	Category: PqoS	Priority: Recommended
Description: The OTT service layer (e.g., AF) shall generate predictive analytics notifications based on service layer performance metrics. To this end, the 5GS shall expose DN performance monitoring information to consuming entities e.g., AFs.		

ID: REQ-F-Task3.3-INTER-3	Name: Support for ML model training	
Type: FR	Category: PqoS	Priority: Recommended
Description: The 5GS shall support the operation of a MLTF NWDAF instance which shall support in turn the training process of the ML models required for PQoS notifications.		

3.2.7 ATSSS-like multi-connectivity

ID: REQ-F-Task3.3-INTER-4	Name: Multi-connectivity mode at the UE through vertical signalling	
Type: FR	Category: Multi-Connectivity	Priority: Recommended
Description: Notification on the multi-connectivity mode shall be sent through NEF to the VS AF		

ID: REQ-F-Task3.3-INTER-5	Name: Non-3GPP access	
Type: FR	Category: Multi-Connectivity	Priority: Recommended
Description: Support for non-3GPP access to the Core Network		

ID: REQ-F-Task3.3-INTER-6	Name: MP-QUIC support at UE and in network	
Type: FR	Category: Multi-Connectivity	Priority: Recommended
Description: The UE and the network should support MPQUIC steering functionality		

ID: REQ-F-Task3.3-INTER-7	Name: Network data monitoring at UE and in network	
Type: FR	Category: Multi-Connectivity	Priority: Recommended
Description: UE should be able to perform performance measurements to decide how to distribute traffic over 3GPP access and non-3GPP access		

4 Architecture enablers and specifications for CAM

The process of architecture definition should commence early in the overall system development lifecycle. This is especially relevant for the ENVELOPE project, which aims to deliver a Beyond 5G system supporting CAM applications and experimentation. The process begins with discussions regarding feasibility, efficiency, challenges, costs, and other factors, typically conducted concurrently with system analysis and requirements definition activities. This results in a comprehensive set of requirements, culminating in an architecture that meets these needs. The input for the architecture description is derived from identified stakeholders and a set of technical, user, and system requirements, alongside architectural decisions. As architecture evolves through a review process involving further architectural decisions, requirements are likely to be updated. A detailed architecture description will provide technical specifications and guide the development process towards the realization of the system architecture.

The ISO/IEC/IEEE 42010 standard, first published in 2011 under the title "Systems and Software Engineering – Architecture Description," offers valuable guidelines by defining considerations for creating an architecture description, although it does not mandate a specific process. The standard provides a conceptual model for architecture description and best practices for defining an efficient architecture. The ENVELOPE reference architecture is based on the methodology, principles, and best practices outlined in the latest update of the standard, issued in 2022. [30]

According to the standard, a system's architecture is defined as "fundamental concepts or properties of a system in its environment, embodied in its elements, relationships, and in the principles of its design and evolution." An Architecture Description (AD) is used to articulate the architecture of a system. Stakeholders have interests in a system, referred to as concerns, which can span a wide range of areas, including technical, personal, developmental, technological, business, operational, organizational, political, economic, legal, regulatory, ecological, and social influences. It is important to note that concerns and requirements are not synonymous; a concern is an area of interest, such as system reliability for some stakeholders.

This section presents the ENVELOPE reference architecture. The main distinction between software architecture and reference architecture is that the former is a design solution for a specific software system, while the latter offers a high-level design solution for a class of similar software systems within a domain. Reference architecture is more abstract and must be instantiated and configured to address the specificities of the software being developed. Such instantiations will be integrated, tested and demonstrated in the various ENVELOPE pilot use cases in WP4 and WP6.

The main objective of ENVELOPE is to advance and transform the reference 5G advanced (B5GS) architecture to be vertical-oriented, by incorporating the necessary interfaces that expose network capabilities to verticals, provide vertical-information to the network and enable verticals to dynamically request and modify certain network aspects in an open, transparent and easy to use semi-automated way. Although basically motivated by and focused on the CAM vertical for the validation of the proposed framework, the resulting developments are expected to be reusable by any vertical and the ENVELOPE architecture targets to serve as an “envelope” that can cover, accommodate and support any type of vertical services.

In order to achieve the objectives set for the ENVELOPE architecture and meet the challenging B5GS requirements of automation and improved user experience via tighter integration of network and service information domains, key innovative technologies, such as mobile edge computing

(MEC), zero touch management, multi-connectivity and PQoS, get into the spotlight. It becomes fundamental for the definition of an all-encompassing architecture to identify and appropriately integrate these innovations –"*enablers*" in the context of the project- shaping a comprehensive end-to-end "*Facility*" where the advanced ENVELOPE UCs and open call projects can be evaluated and demonstrated.

The following subsections start with the overview of the overarching high-level architecture of the envisioned ENVELOPE platform, bringing all components together in a systematic manner, and continue with the more detailed analysis of the key enablers to be integrated, so that to provide a clear understanding of the ENVELOPE developments to be implemented and validated in the project's trial sites.

4.1 ENVELOPE Platform Architecture

A precise and coherent architecture deriving from the objectives set and addressing the requirements collected is of vital importance to ensure the appropriate implementation of the ambitions set. At the same time, understanding the diversity of demands per CAM use case, reflected through the prioritization of requirements per use case as seen in Section 3.3, a modular and layered approach is necessary to achieve an all-encompassing, yet flexible architecture that can be effectively mapped to the underlying system and services capabilities per target trial site. The proposed architecture can be seen in Figure 2, which outlines distinct layers to simplify the deployment process.

The key components of the architecture comprise the following:

ENVELOPE Enablers: A fundamental concept in the ENVELOPE architecture is the layer of the ENVELOPE enablers, which offer important features for testing and optimizing CAM applications. These enablers span from 5GS system functionalities, such as Dynamic Slicing and Session & Service Continuity, to more innovative technologies like ATSSS Multi-Connectivity, MEC Migration, Predictive-QoS, and Zero-touch Management. An ENVELOPE enabler is a software component (or a set of such components) designed to interact with the B5G network's control plane by utilizing exposed Network APIs, such as Northbound APIs of the 5G core and/or MEC APIs, in a standardized and trustworthy manner to support services for vertical industries.

A central concept of ENVELOPE Enablers is their role as software components that extend the openness capabilities of 5G networks. By adhering to CAPIF standards, these enablers ensure reliable integration within mobile networks and enable the creation of new services tailored for vertical industries. Specifically, ENVELOPE Enablers communicate with the network via exposed southbound APIs such as:

- Far-Edge APIs: Linking to Radio Access Networks (RAN) and terminal devices.
- Edge APIs: Providing access to Edge User Plane Functions (UPFs).
- Beyond 5G Network APIs: Facilitating interactions with the B5G core.

An ENVELOPE Enabler is intended to support vertical CAM applications, by exposing APIs, which are termed as ENVELOPE APIs. Third parties, including vertical industries, can develop ENVELOPE Enablers that generate new functional features by leveraging 3GPP APIs as well as other telecommunications assets, including business support system (BSS) APIs, such as service orchestration APIs.

CAM Applications: On top of the ENVELOPE enablers, the CAM Applications are positioned, that can support different “use cases”. CAM Applications are vertical providers and businesses that are interested in exploiting and bringing the new functionality provided by the beyond 5G infrastructure of ENVELOPE, so that it can be used in order to improve their new or existing applications (CAM Applications) by consuming the functionality offered by the ENVELOPE Enablers. Although the main target is the CAM sector, ENVELOPE also seeks, through the open calls, impact on other 5G-enabled vertical industries from the wider ecosystem of the connected vehicles that could be benefited from the ENVELOPE ecosystem.

ENEVELOPE APIs: In between the ENVELOPE enablers and the CAM applications, and fundamental to the architecture of the project, are the ENVELOPE APIs . The ENVELOPE APIs aim to simplify and abstract the interactions of the vertical service with the 5GS acting as an intermediate translation layer that converts the complicated 5GS interfaces and services into easy to consume interfaces accessible by the vertical domain. These APIs allow the applications to request and test features such as dynamic slicing, predictive QoS, ATSSS Multi-connectivity and several others.

Network and the infrastructure layers: Three different segments can be identified in these layers: cloud, edge, and far edge segments. The cloud segment comprises the B5G core system and the cloud resources. The edge segment is made by the local DN and the Edge UPF at the network layer, while the edge infrastructure layer provides the edge computing resources on which to deploy applications. The far edge segment provides the gNB at the network layer, while UEs (e.g. OBU) constitute the infrastructure layer on which to deploy applications, typically, but not necessarily, the client side.

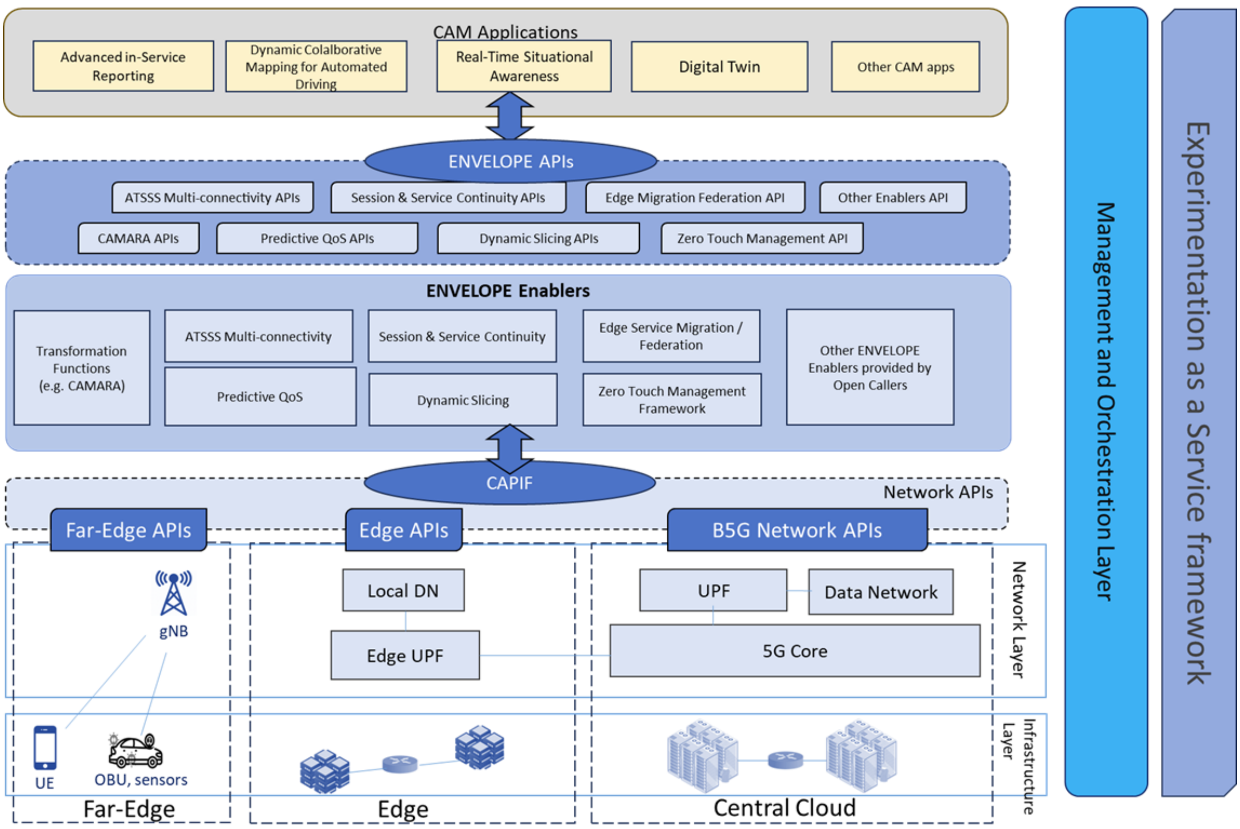

Figure 2 – ENVELOPE Architecture

While the horizontal layers of the architecture address the building blocks of the platform that need to be implemented by the trial sites so that to offer the ENVELOPE services, the vertical order addresses an equally important dimension, the experimentation facilitation. The experimentation-as-a-service module formalises the way that the experimenter can interact with the platform so that to create, execute and evaluate the experiments planned in a systematic manner. As a matter of best-practise any provisioning or configuration action deemed necessary for the experimentation process needs to be managed by the management and orchestration layer of each platform.

4.2 Experimentation as a service

The ENVELOPE project foresees providing an Experimentation as a Service (EaaS) module that has to be used by the experimenter to interact with the ENVELOPE Platform to manage all the aspects related to the experimentation.

The high-level architecture of the EaaS module is reported in Figure 3.

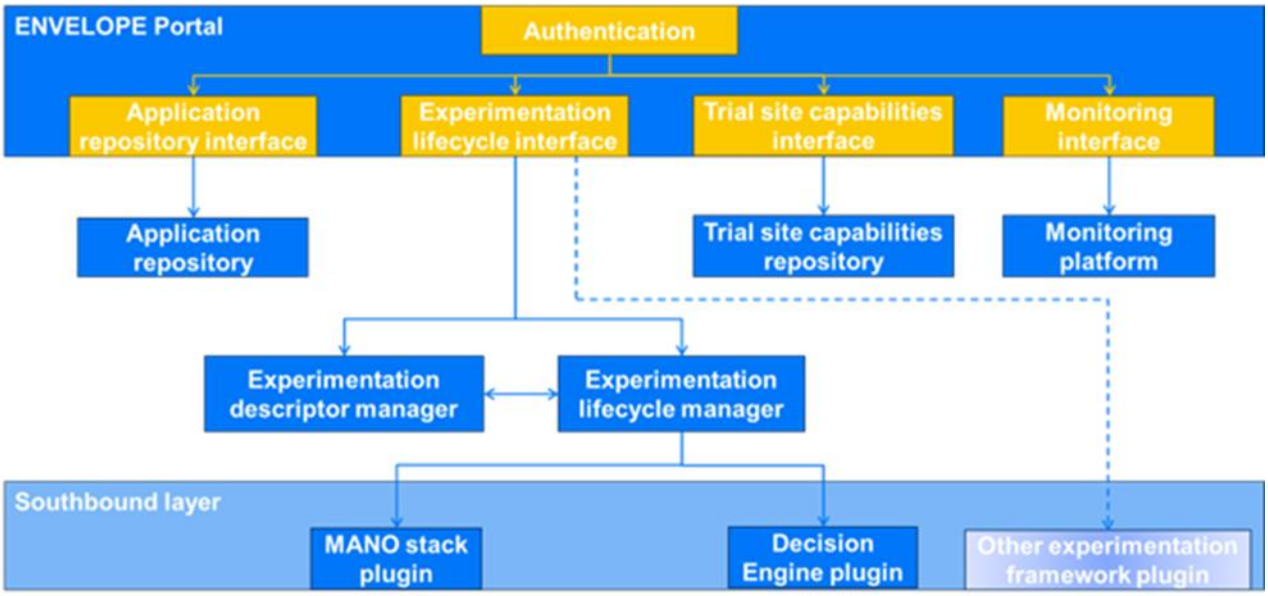

Figure 3 – High-level architecture of the EaaS module

The ENVELOPE Portal is the single point of access of the experimenter towards the ENVELOPE Platform that is deployed in each trial site. The experimenter must first authenticate at the ENVELOPE Portal to access the interfaces towards the ENVELOPE Platform.

The interfaces that are made available to an experimenter are the following ones:

- **Application repository interface**: it allows the experimenter to manage all aspects related to application onboarding (e.g., upload, update, delete).
- **Experimentation lifecycle interface**: it exposes the endpoints for managing experimentation (e.g., start and stop an experiment).
- **Trial site capabilities interface**: the experimenter can access the information describing the capabilities of the trial site.
- **Monitoring interface**: the experiment's metrics and real-time monitoring can be observed by the experimenter through this interface.

The interfaces will allow the experimenter to deal with the corresponding back-end modules using a graphical interaction where possible.

The back-end modules of the ENVELOPE Portal are:

- **Application repository**: it stores the applications that can be used to compose a vertical service to be experimented with. It contains the necessary artifacts to deploy the application (OCI-images, qcow2 or iso images) that will be then downloaded by the Management and Orchestration Layer to deploy the services. It acts as a registry for the experimenters to upload their components.
- **Experimentation descriptor manager**: it is responsible for processing the experimentation descriptor that is provided by the experimenter. It analyses the descriptor for checking syntax errors and any other issues that can prevent running the experiment described.
- **Experimentation lifecycle manager**: it reacts to the triggers received by the experimenter for starting or stopping an experiment. It interacts with the “Experimentation descriptor manager” to retrieve the details of the experimentation and it forwards the needed information to the Southbound layer’s plugins.
- **Southbound layer plugins**: these plugins are responsible for interacting with the modules of the B5GS and of the ENVELOPE Platform which manage the network and computing resources and the service orchestration. The number and the type of plugins are strictly related to which Management and Orchestration modules are deployed in the trial site. It is possible to exploit optionally other experimentation frameworks by deploying ad-hoc plugins for the purpose. In this case, the ENVELOPE Experimentation lifecycle manager is not used, as the functionalities of the other framework for the experimentation lifecycle are exploited e.g., 6G-SANDBOX framework.[31]
- **Trial site capabilities repository**: it stores the information about the capabilities of the trial site. The capabilities include, but are not limited to, the ENVELOPE APIs supported by the trial site, the ENVELOPE Enablers available, and the computing resources available at the far edge, edge and cloud infrastructure segments.
- **Monitoring platform**: it is responsible for collecting the metrics, logs and other data that the experimenter needs for monitoring in real-time the experimentation and for evaluating the outcomes of the experiment.

4.3 Enablers for Management and Orchestration

4.3.1 Management and Orchestration Layer

The Management and Orchestration Layer enables efficient and dynamic service delivery. The layer integrates multiple components to provide a comprehensive framework for managing and orchestrating computing and network resources, as well as the services running on top of virtualized infrastructures, enabling automated deployment, scaling and optimization of applications.

Based on the requirements collected and described in Section 3, the ENVELOPE Management and Orchestration Layer is the modular system to ensure that the network and the computing infrastructures are able to meet the application requirements when the user require an application to be instantiated to run an experiment through the ENVELOPE Experimentation as a Service Module.

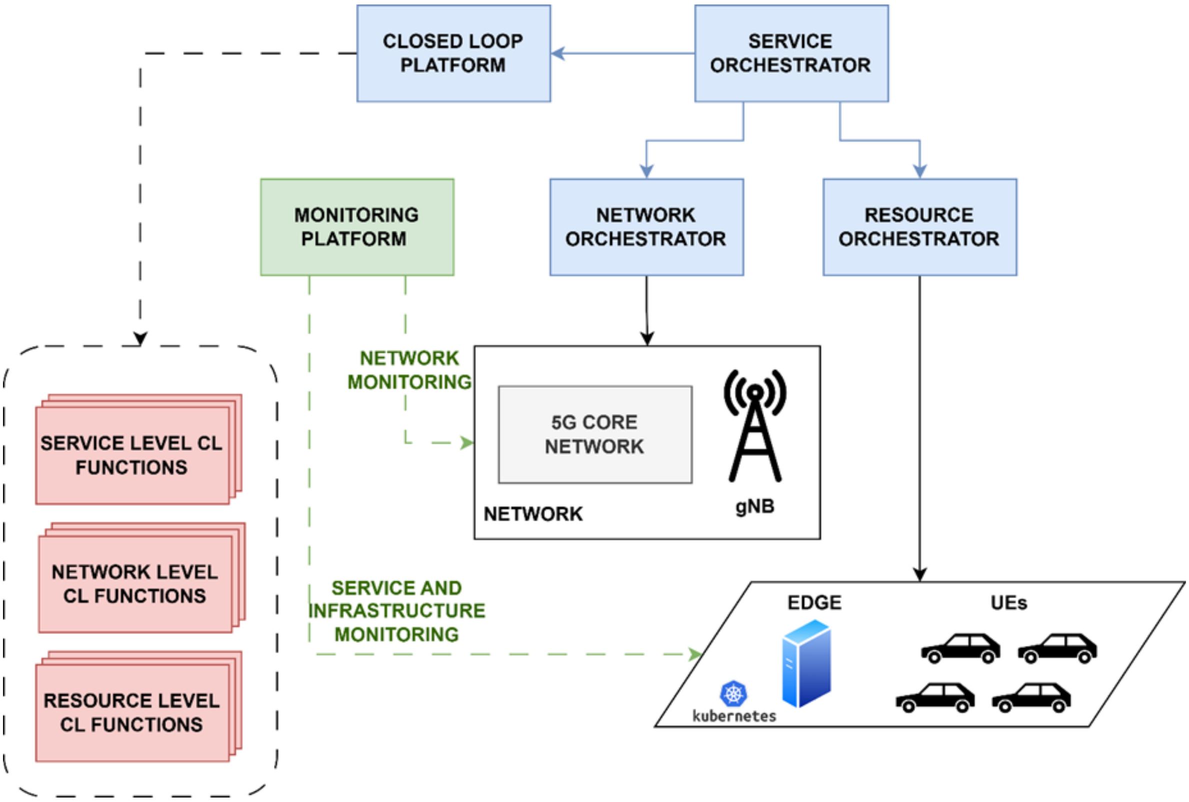

Figure 4 – Management and Orchestration Layer

Figure 4 shows the Management and Orchestration high level architecture. The layer is composed of integrated components that cooperates to enable the automated operation of 5G networks and services:

- **Service Orchestrator:** implements all the functionalities to manage Vertical & Network Service lifecycle, by receiving command from its northbound interfaces. It includes components to manage service predefined descriptor and artifacts through catalogues and artifact registries (like OCI-based registries), as well as a Service Lifecycle component to instantiate, (re-)configure and terminate applications.
- **Resource Orchestrator:** manages computing resources across the Extreme Edge/Edge/Cloud continuum. It operates at the infrastructure level, provisioning the required compute, storage resources across distributed data centers, edge nodes, and cloud environments. It interacts with platforms like Kubernetes to manage virtualized resources dynamically based on the demands received from the Service Orchestration layer.
- **Network Orchestrator:** is the component in charge of coordinating network resources, by deploying, configuring and executing runtime operations on 5G Network Slices. It ensures that the allocated resources are properly aligned with the network requirements of the service. Components such as the Network Slice Management Function (NSMF) and Network Slice Subnet Management Function (NSSMF) enable fine-grained control of network slices.
- **Closed-Loop Platform:** enable real-time automation and optimization at infrastructure, network and service level. It enables the possibility to instantiate and orchestrate Closed Loop

Functions that collect feedback from the Monitoring Platform to detect anomalies, predict resource demands, and trigger corrective actions like scaling, healing, or reconfiguration.

- **Monitoring Platform:** provide the data foundation for the Management and Orchestration layer, offering visibility into resource usage, service performance, and network health. This platform continuously feeds metrics to the Closed-Loop Systems and other components, ensuring a data-driven approach to decision-making and maintaining operational efficiency. At the experimentation level, the Monitoring Platform is in charge of collecting KPIs to evaluate the performances of the applications.

The interaction between external user (experimenters) and the platform starts with the service design and on-boarding phase, where application components (i.e. virtualized software modules) should be uploaded in the platform to enable easy and standardized deployment through the Management and Orchestration system. Figure 5 depicts the main steps to on-board application packages into the platform, where the Application Registry is realised by an Open Container Initiative (OCI) compliant Artifact Registry, which are are designed to store and manage container images, such as Docker images, in a standardized way.

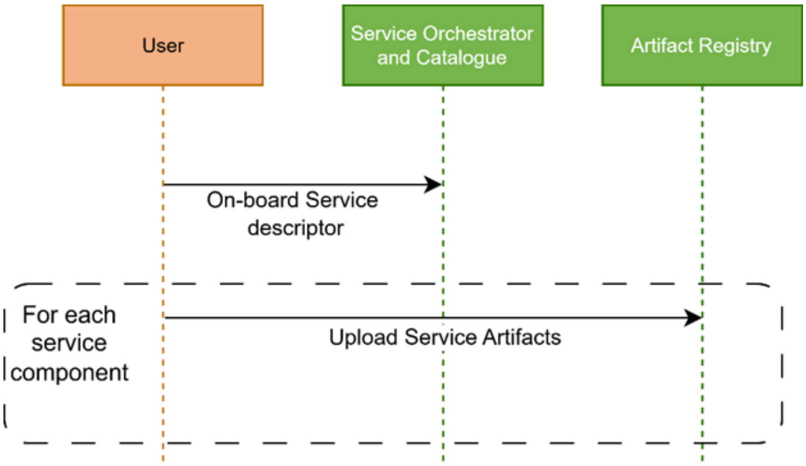

Figure 5 – Service Design and on-board workflow

Vertical services are commonly composed by several application components, each of them implementing an atomic functionality of the overall application. The user (experimenter) will upload a Vertical Service descriptor in the Vertical Service catalogue to describe all the requirements to deploy the application (like computing requirements, URL to component artifacts, exposed ports etc) as well as the application components that the Vertical Service is composed by. Once the Vertical Service descriptor is uploaded, the application component artifacts should be uploaded in the Artifact Registry (like OCI-Registry) if not publicly available. OCI-compliant registries ensure that container images can be easily shared, deployed, and managed across different environments and platforms.

Once service descriptors and artifacts are available to the Management and Orchestration layer, the Experimentation as a Service platform is able to instantiate services and configure the network, as described in the workflow diagram at Figure 6.

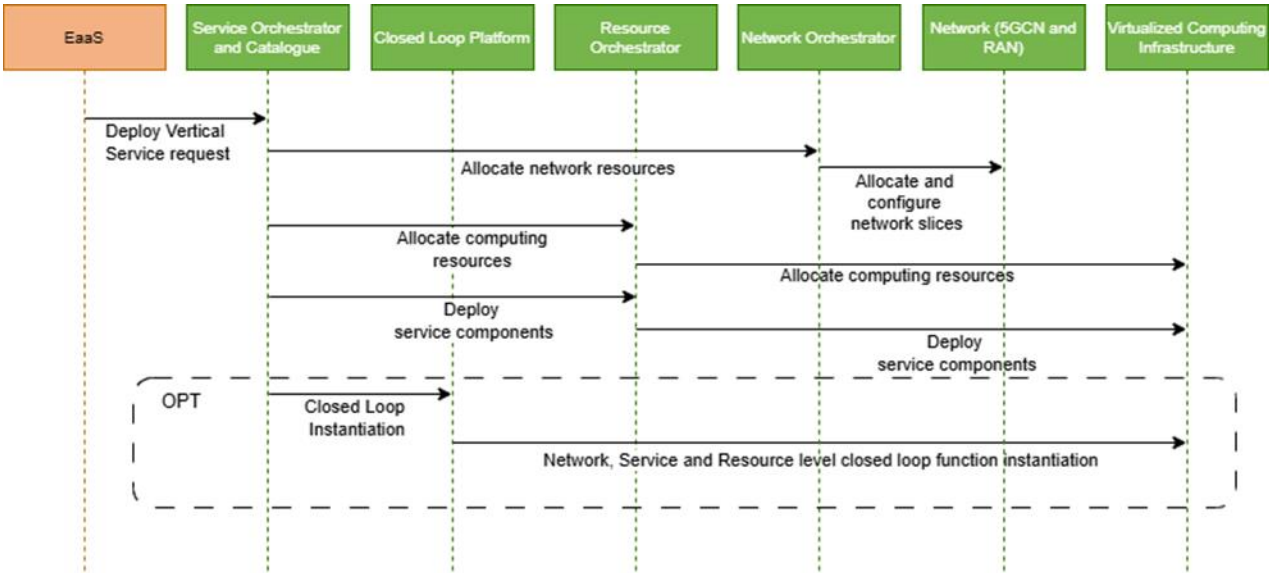

Figure 6 – Service instantiation and Network Configuration workflow diagram

This is achieved through the following steps:

- **Step 0)** Experiment design (not depicted in the Figure): Experimenters should describe the experiment by uploading an Experiment Descriptor or by interacting with EaaS through the interactive GUI.
- **Step 1)** EaaS requests the instantiation of a Vertical Service: once the experiment is defined, EaaS triggers the deployment of a Vertical Service to the Management and Orchestration Layer's entrypoint component (the Service Orchestrator), which is the component in charge of translating the request into the actions to be executed into the computing infrastructure and into the network.
- **Step 2)** Allocation of network resources: if the experiment requires network resources to be allocated, a dedicated action (driven by the Network Orchestrator) ensures that proper network configuration is executed for each segment. This includes the instantiation or configuration of 5G network slices in in the 5G core network and/or the proper configuration of the 5G radio access network (RAN) to ensure the desired quality of service.
- **Step 3)** Computing resources allocation: application components run on top of virtualized computing infrastructures. To ensure that the application performances meet the required quality of service, the computing resources (like virtual CPUs, memory, and disk storage) should be pre-allocated into the computing infrastructure. This is achieved by requesting the allocation of such resources by the Resource Orchestrator.
- **Step 4)** Service Instantiation: once the network and computing resources are allocated, the Service Orchestrator triggers the deployment of the application components, to actually instantiate them into the computing infrastructure.
- **Step 5)** (OPT) Closed Loop Instantiation: if the Service requires the instantiation of a closed loop, the Service Orchestrator delegates the Closed Loop (CL) Platform to orchestrate the functions that are part of the CL.

4.3.2 Native AI zero touch management

Part of the Zero Touch ecosystem of the project are the Intent Engine and the Intent Handlers (one per application). The Intent Engine module enables users to express service requests in high-level intent representations (following TMF 921 standard – part of the TMF IDAN Catalyst project [27]) streamlining network operations and translating user intent into actionable tasks. The actions based on the Intents are handled by the respective Intent Handler. In our case there will be Decision Engines (RL agents – one per application) that will control specific parameters related to a CAM application (e.g. Anomaly Detection Sensitivity), dynamically. Each Intent Handler will be part of the respective Decision Engine. As mentioned, the latter will perform some action – related to a specific CAM application. The Decision Engine will be integrated with the Management and Orchestration modules presented above (see Figure 7) and it will be managed by the Closed Loop Governance module. Thus, the workflow is: the user uses a frontend component to enter high-level intent representations. Then conflict resolution takes place. Finally, the intents are translated into a format that the Management and Orchestration solution (presented above) understands.

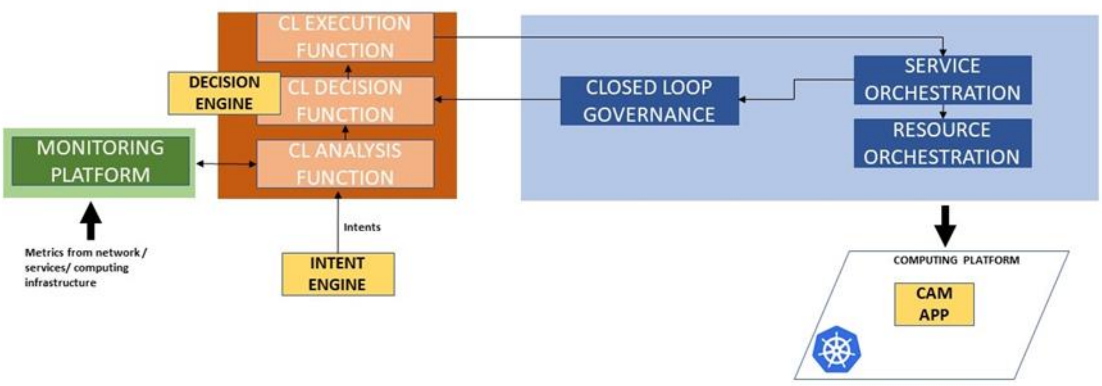

Figure 7 – Integration of the Zero Touch Management modules with the Intent and Decision Engines.

4.4 Network APIs

In 5G systems, the concept of trusted and untrusted domains is crucial for ensuring secure and efficient communication. When a 5G system exposes its network APIs, it must differentiate between these domains to maintain security and integrity.

A **trusted domain** is a network or environment that is considered secure and under the control of the network operator. These domains have established trust relationships with the 5G core network and are allowed to access network resources and services without additional security checks. Examples of trusted domains include the operator's own network infrastructure and trusted partners.

An **untrusted domain**, on the other hand, is a network or environment that is not under the control of the network operator and is considered less secure. These domains require additional security measures to access network resources and services. Examples of untrusted domains include public Wi-Fi networks and third-party applications.

The differentiation between trusted and untrusted domains is achieved through the use of security policies and mechanisms such as authentication, authorization, and encryption. When a network API is accessed, the system checks the domain's trust status and applies the appropriate security measures.

4.4.1 Exposure for untrusted domain

Within 5GS, different 5G network functions are appropriate for configuring a particular monitoring event, detecting the event, and reporting to an authorised external application. UE's mobility management context, including UE position, reachability, roaming status, and loss of connectivity, can be exposed using the monitoring feature (i.e., Event Monitoring API). The section below aims to present and describe the flow diagrams (control plane signalling) of the relevant Northbound APIs.

4.4.1.1 Event Monitoring API - Location Reporting Event

The event is detected based on the event reporting information parameters (i.e., LOCATION_REPORTING) included in the Monitoring request. These parameters include details like one-time reporting, the maximum number of reports, maximum reporting duration.

Location Reporting Options:

- One-time Reporting: The event reporting provides either the current location or the last known location of the UE. If the immediate reporting flag is set, AMF reports the Last Known Location immediately. If the immediate reporting flag is not set, AMF reports the UE's current location. If AMF does not have the current location at the requested granularity, it retrieves the information via NG-RAN Location reporting procedure.
- Continuous Location Reporting: For continuous reporting, the serving node sends notifications every time a location change is detected, with the granularity depending on the accepted accuracy of location. In the context of the project, the granularity is per cell level.

The location request precision is lower or equal to cell level, thus the consuming function (i.e., AF) is notified for location updates triggered when a specific UE changes cell. The following Figure below, is conducted by the overall information flow described in TS 23.502. The NFs being involved in the process are AF, NEF, UDM and AMF. Moreover, the APIs for the interaction between NFs are identified (adapted from TS 29.522).

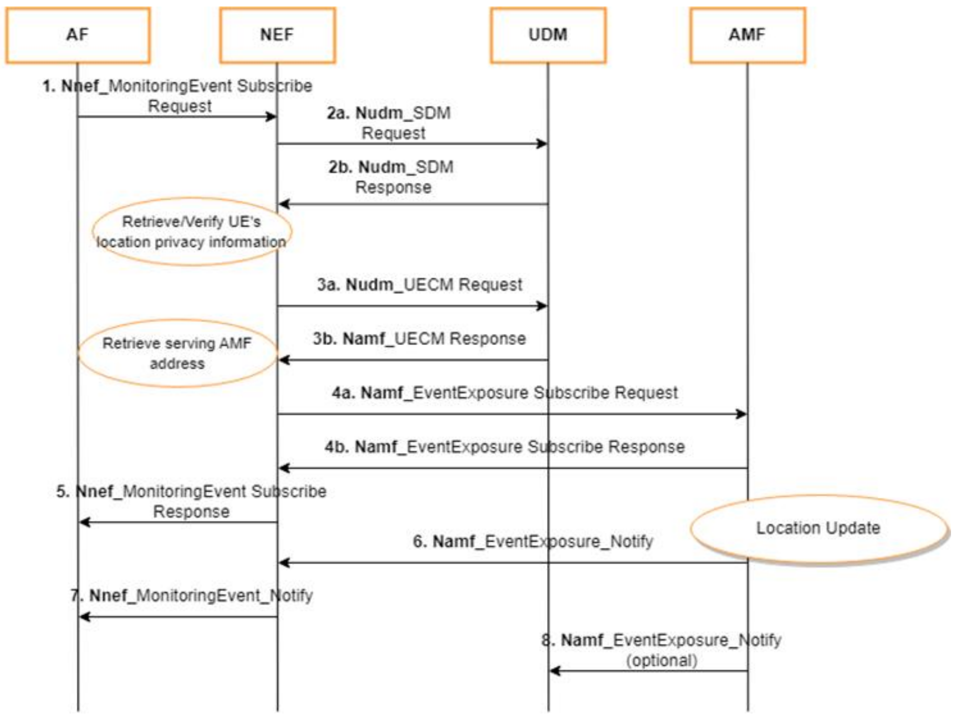

Figure 8 – Information flow for the Location reporting event

Steps:

1. AF requests a subscription through Monitoring Event API provided by NEF for LOCATION_REPORTING events.
2. NEF retrieves the UE location privacy information from the UDM using the **Nudm_SDM** service (for more details refer to TS 29.503).
3. If the privacy setting is verified, then NEF interacts again with UDM to discover the serving AMF address by invoking the **Nudm_UECM** service.
4. After receiving the service AMF address, NEF interacts with the AMF and subscribes for the location_report event through **Namf_EventExposure** service.
5. NEF informs the AF that the subscription was successfully created.
6. When the event occurs in the 5GS the AMF informs the NEF with a callback notification
7. NEF forwards the notification to the AF.
8. (Optional) If the AMF detects a subscription change related to the LOCATION_REPORTING event, it sends the event report, by means of Namf_EventExposure_Notify message to the UDM.

In the Table below, a summary of the APIs used in the above procedure is presented.

NF provider	NF consumer	Service	API	3GPP Document / GitHub link
NEF	**AF**	MonitoringEvent API	`POST /{scsAsID}/subscriptions`	TS 29.122/29.522 – link
UDM	**NEF**	SDM API	`GET /{ueId}/lcs-privacy-data`	TS 29.503 – link
UDM	**NEF**	UECM API	`GET /{ueId}/registrations/amf-3gpp-access`	TS 29.503 – link

			or `GET /{ueId}/registrations`	
AMF	**NEF**	EventExposure API	`POST /subscriptions`	TS 29.518 – link

Table 1 APIs used in the Location reporting event

4.4.1.2 AsSessionWithQoS API

Applications can utilize AsSessionWithQoS API to ensure better service experience and avoid service interruption which may be provoked due to unexpected QoS downgrades. 5GS comprises a complex QoS model where different networks function from the 5GC and the RAN interchange information to make the necessary adaptations for the best possible end-to-end connectivity (i.e., 5GS provides connectivity by establishing a session between UE and the data network).

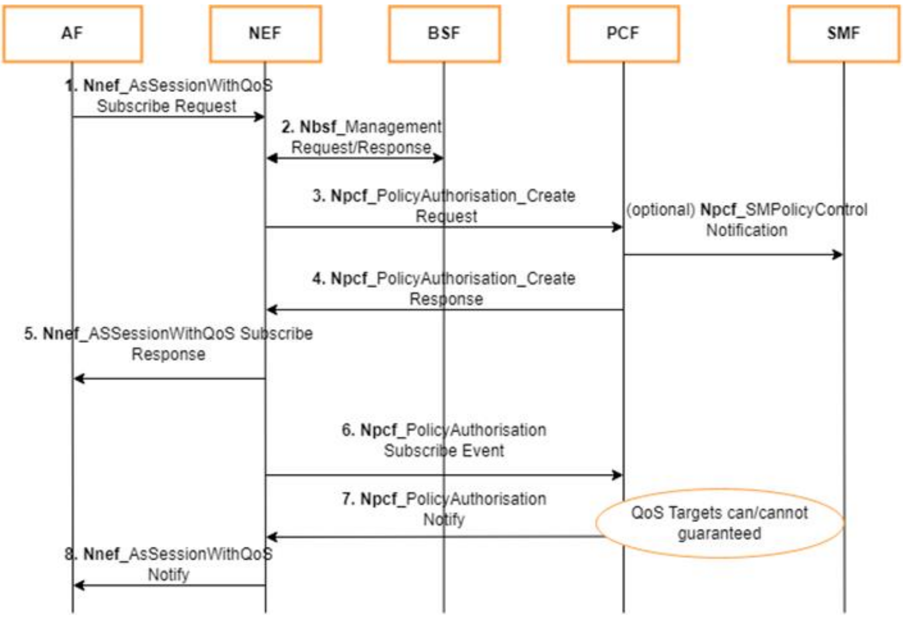

Figure 9 – Information flow for the AsSessionwithQoS

1. The AF sends a request to subscribe for QoS Flow using the QoS Reference attribute through **Nnef_AsSessionWithQoS**.
2. NEF uses the UE address to discover the PCF from the BSF through **Nbsf_management.**
3. After discovering the PCF, NEF forwards the received parameters from the AF to PCF through **Npcf_PolicyAuthorisation**
4. The PCF derives the required QoS parameters of the PCC rule based on the information provided by the NEF and determines whether this QoS is allowed (according to the PCF configuration) and notifies the result to the NEF.
5. If the PCF determines that the SMF needs updated policy information, the PCF issues a **Npcf_SMPolicyControl_Notification** request with updated policy information about the PDU Session.
5. The NEF sends a response message (Transaction Reference ID, Result) to the AF. The result indicates whether the request is granted or not.

6. NEF subscribes for the event (QOS_NOTIFY) (i.e., QoS targets can no longer/ can again be fulfilled) [23503] to the PCF.
7. When the event condition is met, e.g., that the establishment of the transmission resources corresponding to the QoS update succeeded or failed, the PCF sends **Npcf_PolicyAuthorization_Notify** message to the NEF notifying about the event.
8. NEF forwards the notification to the AF.

In the Table below, a summary of the APIs used in the above procedure is presented.

NF provider	NF consumer	Service	API	3GPP Document / GitHub link
NEF	AF	AsSessionWithQoS API	`POST /{scsAsID}/subscriptions`	TS 29.122/29.522 – link
BSF	NEF	Management API	`GET /pcfBindings`	TS 29.521 – link
PCF	NEF	PolicyAuthorization Service API	`POST /app-sessions`	TS 29.514 – link
PCF	SMF	SMPolicyControl API	`POST /callback`	TS 29.512 – link
PCF	NEF	PolicyAuthorization Service API	`POST /app-sessions/{appSessionId}/events-subscription`	TS 29.514 – link

4.4.1.3 AF Traffic Influence API

The Traffic Influence API in 5G is a pivotal element for the management and optimization of network traffic. This API facilitates the interaction between applications and the 5G network, enabling the influence of traffic routing and quality of service (QoS) policies. Situated within the control plane of the 5G Service-Based Architecture (SBA), the AF communicates with other network functions such as the Policy Control Function (PCF) and the Network Exposure Function (NEF) as the following figure depicts. Through the Traffic Influence API, applications can request specific traffic handling rules, including the prioritization of certain data types and the assurance of low latency for critical services. This functionality is indispensable for applications demanding high performance and reliability, such as real-time video streaming, online gaming, and industrial IoT applications.

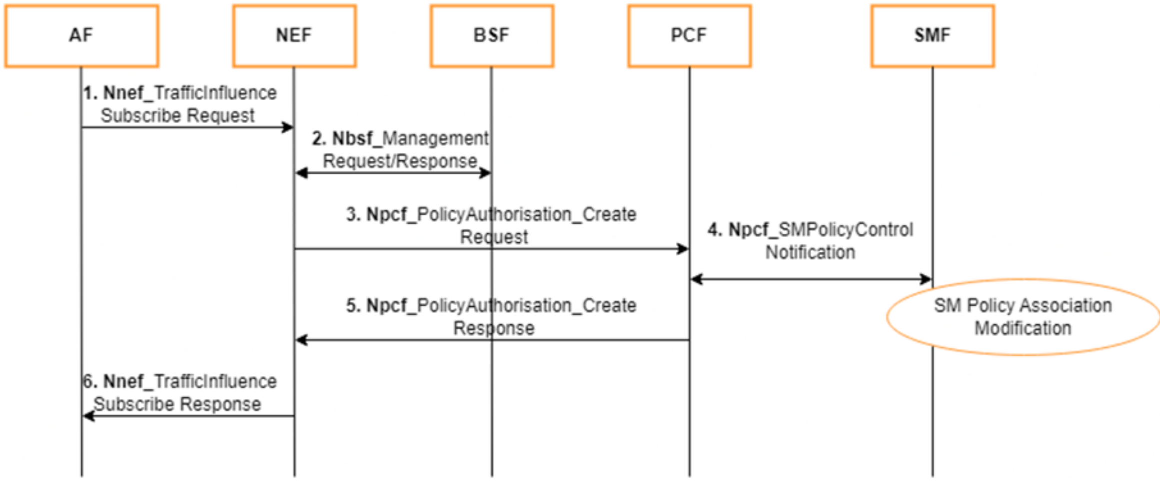

Figure 10 – Information Flow of AF Traffic Influence API

More specifically:

1. The AF sends a request to subscribe for influencing traffic routing to a local network, targeting an individual UE address, through **Nnef_TrafficInf**.
2. NEF uses the UE address to discover the PCF from the BSF through **Nbsf_management.**
3. After discovering the PCF, NEF forwards the received parameters from the AF to PCF through **Npcf_PolicyAuthorisation**
4. The PCF derives the required traffic routing parameters based on the information provided by the NEF and the PCF updates the SMF with corresponding new PCC rule(s) with PCF initiated **SM Policy Association Modification** procedure as described in clause 4.16.5.2 of 3GPP TS 23.502.
5. PCF returns the response to the NEF.
6. The NEF sends a response message to the AF. The result indicates whether the request is granted or not.

In the Table below, a summary of the APIs used in the above procedure is presented.

NF provider	NF consumer	Service	API	3GPP Document / GitHub link
NEF	AF	TrafficInfluence API	`POST /{afId}/subscriptions`	TS 29.122/29.522 – link
BSF	NEF	Management API	`GET /pcfBindings`	TS 29.521 – link
PCF	NEF	PolicyAuthorization Service API	`POST /app-sessions`	TS 29.514 – link
PCF	SMF	SMPolicyControl API	`POST /callback`	TS 29.512 – link

4.4.2 Exposure for trusted domain

The 5GC exposes the N5 reference point, used for the interaction between trusted AFs and the PCF, without NEF mediation. Via N5, external trusted AFs can directly consume the PolicyAuthorization Service described above to configure network parameters and perform end-to-end 5G QoS modification and control, typically to support mission-critical services.

Furthermore, in the particular case of the HPE Aruba Networking Private 5G Core implementation, the core exposes several *northbound RESTful APIs* leverageable by external Operation, Administration, and Monitoring (OAM) systems to carry out automated configuration, performance, and fault management procedures. These APIs follow the OpenAPI 3.0 specification and can be consumed by any client able to produce HTTP requests and interpret HTTP responses, regardless of its programming language. Through the HPE 5GC APIs, the configuration settings of network functions are exposed and can be re-parametrized, fostering system openness and dynamic control. In other words, it is possible for authorized and trusted external parties to programmatically adjust settings, deploy new configurations, provision new users, and manage the network behavior directly through the APIs. In particular, Provisioning APIs are specified in 3GPP TS 29.504 UDR Network Function, that is the network entity that allows storage, retrieval and modification of the subscription data as specified in 3GPP TS 29.505 and policy data as specified in 3GPP TS 29.519. This API can be utilized by a trusted AF or orchestrator to associate and modify the QoS of subscribers. In this framework, the UDR acts as NF Service Producer. It provides the Data Repository service to an NF service consumer.

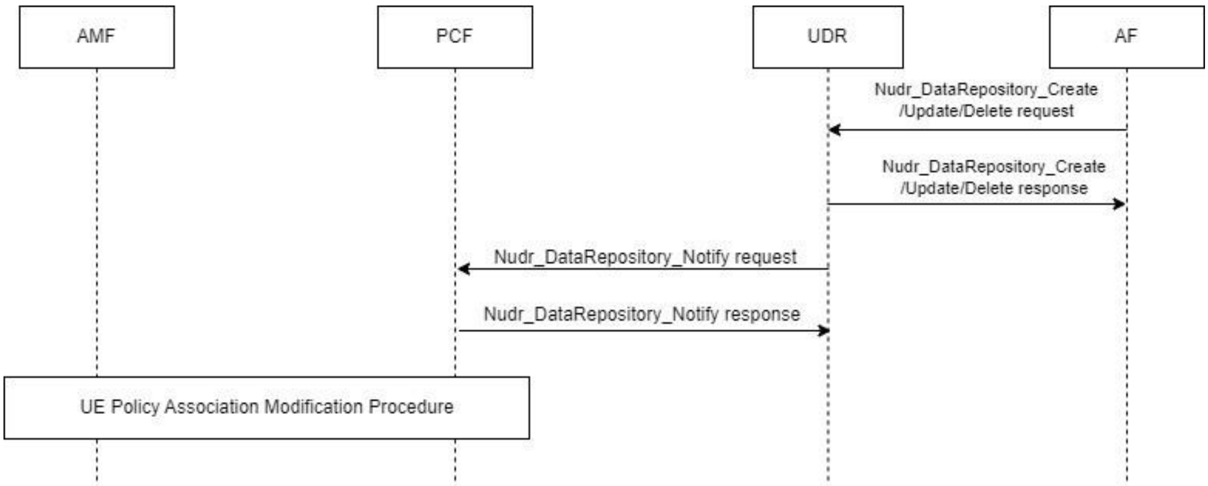

Figure 11 – High-level information flow for user data provisioning via API

As summarized in the figure above, to update a user's QoS parameters via user data re-provisioning, the update service operation is used by the NF service consumer to update the data stored in the UDR. The following procedures using the Update service operation are supported: data update using PATCH or PUT.

NF provider	NF consumer	Service	API	3GPP Document / GitHub link

UDR	**AF**	Data repository Create/Update APIs	PUT PATCH {apiRoot}/nudr-dr/ policy-data	TS 29.519
UDR	**PCF**	Subscription to modification of Policy data APIs	POST {apiRoot}/nudr-dr// policy-data/subs-to-notify	TS 29.519

Moreover, it is possible to collect real-time and historical monitoring data or network performance metrics via API. Network function information is collected and exposed using Prometheus. They encompass resource consumption data, 3GPP-related metrics for signaling and traffic data, and monitoring information about the state of the network. Further, the HPE 5GC exposes via API a set of parameters for monitoring its virtual machines, including metrics such as CPU utilization, memory allocation, disk usage, networking data.

We highlight that Transport Layer Security (TLS) certificates protect the access to all exposed APIs from unauthorized entities and prevent data breaches. TLS is a cryptographic protocol that allows server identification and authorization. It guarantees that API calls are transferred to the intended recipient, while impeding data interception.

Considering the urgent need to respond quickly to emergencies, especially in Automated Driving Vehicles (AVs) scenarios, the B5G/6G network must be designed to handle resource allocation tasks efficiently, while guaranteeing the appropriate QoS for critical services. As an example, when an AV is involved in an accident with other AVs (neighbors), it is crucial to prioritize within the network all communication related to safety and to critical message exchange. Another scenario could involve enhancing data sharing for collaborative mapping, necessitating a QoS that supports functionalities like streaming, for instance when work is being conducted on the road.

Focusing on the 5GC, one way to support the needs of different traffic types is to associate them with different network slices. This approach is illustrated in Figure 12. As a reference, let us consider a 5GC architecture with control plane functions shared by two slices in the network and, oppositely, dedicated User Plane Functions (UPFs) for each of the two slices. In routine conditions, “general-purpose” traffic of all users may be let flow through UPF1, with best effort performance; UPF2 may be kept off, for the sake of energy efficiency or simply not to inefficiently occupy an edge infrastructure’s resources with redundant functionalities. Let us make the hypothesis that UPF2 is designed to handle only non-routine high-priority traffic, for example, during AV/car accidents or when collaboration is necessary to update a map. Whenever a critical or privileged communication service needs to be supported, the network can switch on UPF2, and the data traffic of users associated with such services (e.g., AVs) can be shifted to UPF2. In this manner, critical traffic is assigned to dedicated network resources (a UPF within a specific slice), and it is separated from uncontrolled best-effort traffic that may impair the targeted QoS. Notice also that the UPF activation management and the traffic shift among slices can be handled by an orchestrator via 5GC management APIs. For example, in case of a traffic incident, the orchestrator can use 5GC APIs to activate a new UPF and manage the emergency communication slice. Similarly, for map updates, the orchestrator can shift traffic between slices (e.g. after detecting a road closure) to exchange data and ensure the updated map is delivered efficiently.

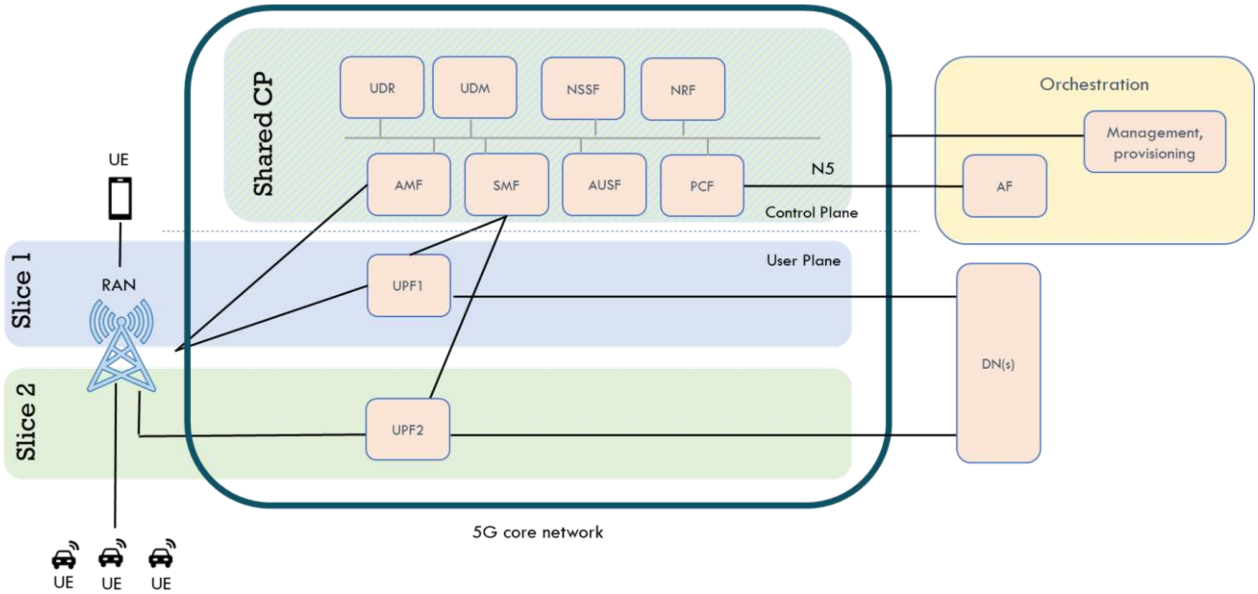

Figure 12 – Slice management for critical service support.

Further, traffic prioritization and QoS differentiation can also be put into practice within the resources of a same slice. In particular, the 5GC can apply QoS policies to traffic of specific applications and services to differentiate them from “general-purpose” traffic. Such rules are considered by the PCF, which plays a crucial role in managing the UEs policies and are forwarded to the Session Management Function (SMF) to create dedicated QoS flows to support the considered application traffic. Such QoS flows are characterized by the QoS parameters specified in the related policies (e.g., 5G QoS Indicators, 5QIs), and such parameters translate into the required QoS and network performance, in terms of prioritization, guaranteed bit rate, minimum bandwidth, etc. Different QoS flows coexist within PDU sessions of a slice. Note that a trusted external AF (and an orchestrator acting as such) can control the application of dedicated policies by interacting with the PCF over the exposed N5 interface.

A high-level flow diagram that summarizes the described interactions between a 5G system and the other components that control a vertical service is shown in the following Figure.

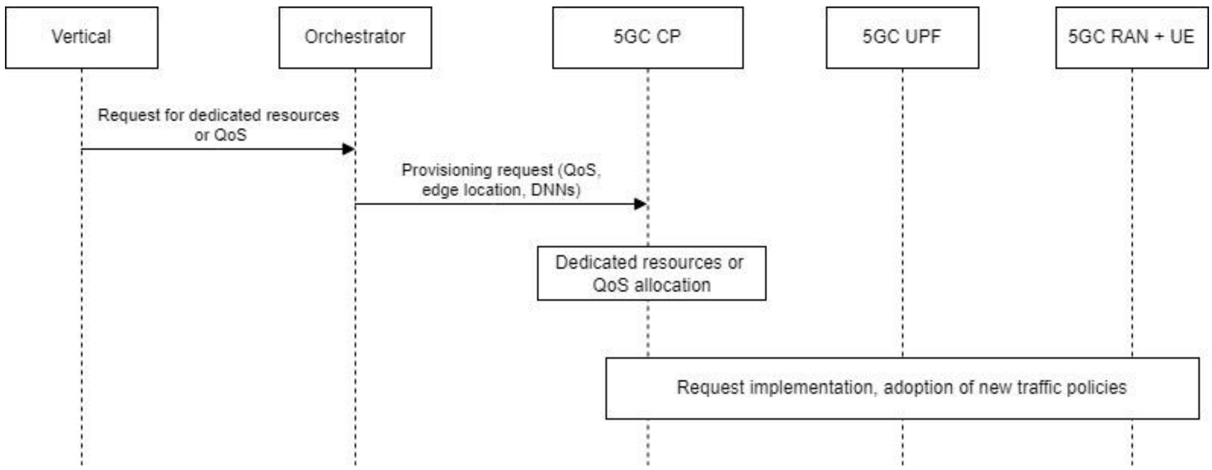

Figure 13 – High-level flow diagram of request for resource re-allocation or QoS modification

4.4.3 Exposure of Network APIs via CAPIF

One of the key aspects in network API exposure is the capability to offer the API services in an interoperable and secure way, while the services are easily discoverable and monitored. In view of that, a standardized framework has been selected, namely the 3GPP Common API Framework (CAPIF) [32]. CAPIF includes common aspects applicable to any northbound service API. As such, it is a complete 3GPP API framework that covers functionality related to on-board and off-board API invokers, registering and releasing APIs that need to be exposed, discovering APIs by third parties, and authorization and authentication.

The CAPIF plays a crucial role in balancing the situation between trusted and untrusted domains in 5G systems by providing a standardized interface for accessing network services. CAPIF ensures that APIs are securely exposed to both trusted and untrusted domains. This means that even third-party applications in untrusted domains can access network services in a controlled and secure manner. CAPIF includes mechanisms for authenticating and authorizing API requests. Trusted domains can use pre-established credentials for easier access, while untrusted domains must go through additional authentication procedures. CAPIF provides services like topology hiding, which ensures that the internal network topology is not exposed to untrusted domains. This helps in maintaining the security and integrity of the network. By using a common API framework, CAPIF allows for scalable and robust communication between the 5G core network and third-party applications, ensuring that the system can handle a large number of requests without compromising security. By implementing these features, CAPIF effectively balances the needs of both trusted and untrusted domains, ensuring secure and efficient communication within the 5G ecosystem.

Three main architectural entities have been defined for CAPIF, namely, the API invoker, the CAPIF Core Function, and the API provider. The API invoker is typically provided by a third-party application and needs to consume APIs from an API provider. The CAPIF core function is the main entity of the CAPIF and it consists of engines that among other capabilities they authenticate the API invoker based on identity and/or other information, authorise API invokers prior to accessing service APIs, onboard/offboard API invokers, monitor service API invocations, and store policy configurations related to CAPIF and service APIs. The API provider is an entity that provides API Exposing, Publishing and Management functions. More precisely, the API exposing function (AEF) is the provider of the service APIs and is also the service communication entry point of the service API to the API invokers; the API publishing function (APF) enables the API provider to publish the service APIs information in order to enable the discovery of service APIs by the API invoker; and, the API management function (AMF) enables the API provider to perform administration of the service APIs. Based on the three fundamental entities that CAPIF architecture has defined, a set of reference points, with associated management APIs, are specified as well. Those interfaces are labelled from CAPIF-1 to CAPIF-5 for API invokers/providers that are inside the same PLMN domain, with the CCF; and from CAPIF-1e to CAPIF-5e for API invokers/providers that are outside the PLMN domain of CCF. For the communication among CCF of the same or different providers the CAPIF-6 and CAPIF-6e reference points are used, respectively. Finally, for the communication among API providers within or outside PLMN domain the CAPIF-7 and CAPIF-7e reference points are used, respectively.

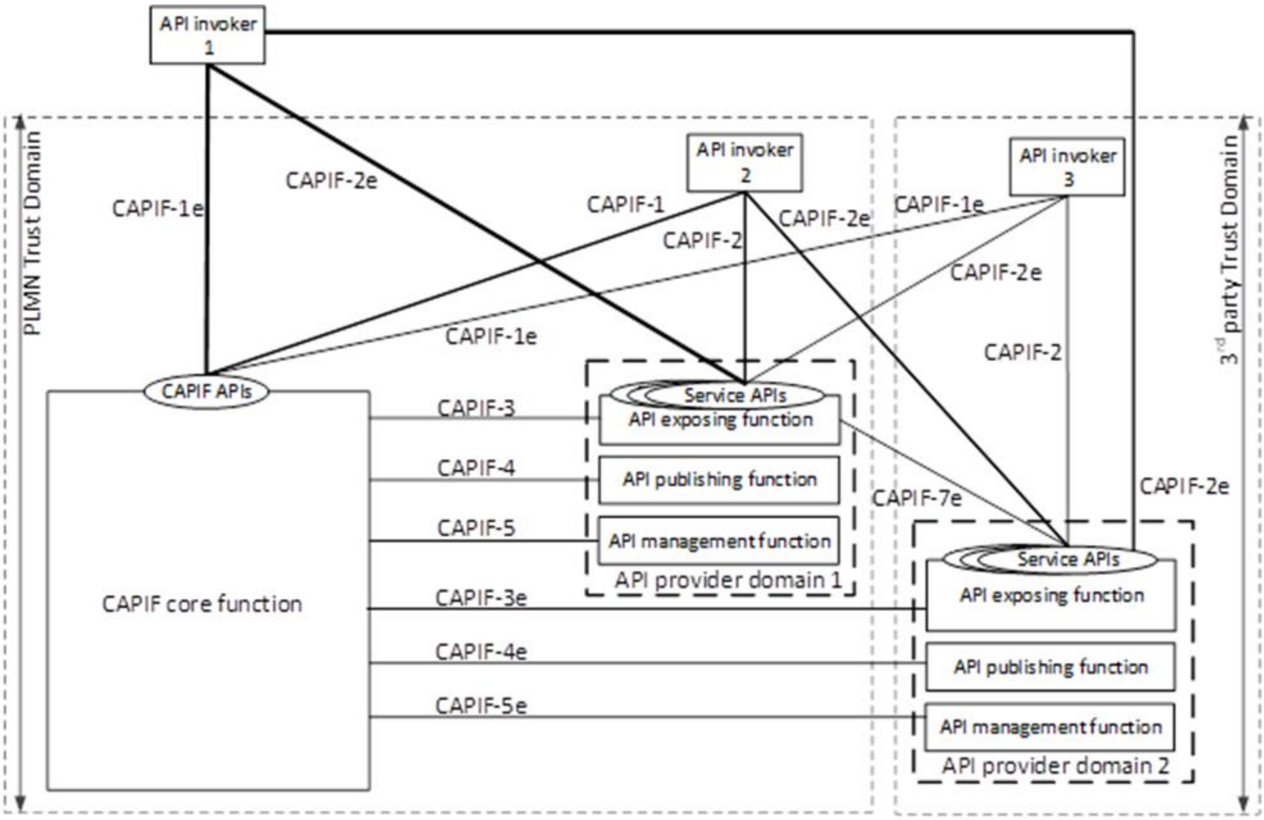

Figure 14 – CAPIF architecture as defined in 3GPP TS 23.222. Main entities and reference points ae depicted.

Considering functionality, any API provider that resides at any domain of the ENVELOPE architecture (Central cloud, edge, Far-edge) can publish its APIs through the CAPIF core function, while any API consumer or invoker shall be onboarded and authenticated by the CAPIF core function to discover and use the published APIs.

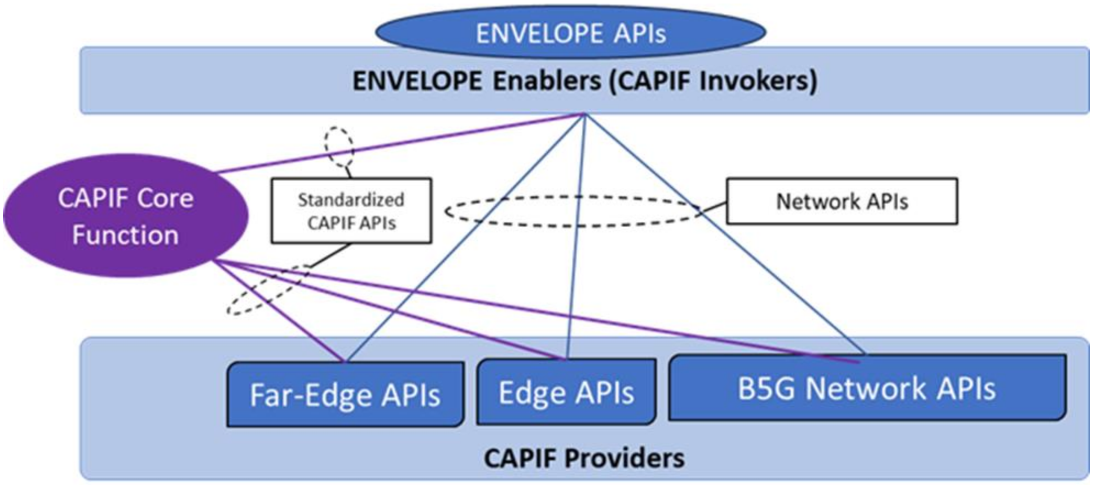

Figure 15 – CAPIF in the ENVELOPE architecture. The CAPIF core function facilitates the exposure of Network APIs by providing security, publishing, discovery and monitoring services.

ENVELOPE will use the OpenCAPIF implementation of CAPIF (as provided by ETSI SDG OCF). The basis will be OpenCAPIF Release 1. In this framework, the ENVELOPE project has already been added in the ETSI SDG OCF research ecosystem[1].

4.5 Enablers for Edge Computing

4.5.1 State-of-the-art overview on edge computing

The past years, most mobile operators have implemented a strategic transition from physical network infrastructure to cloud-based architectures, investing in NFV networks based on cloud and containerised technologies. The 5G SA service-based architecture is paving the way for edge-cloud adoption, relishing performance, and security benefits as well as financial gains through the intelligent use of computational resources. As a result, for the edge-cloud developments in the mobile networks domain a strong technological ecosystem has been developed with the involvement of key players and Standards Development Organisations (SDOs). Practically, as highlighted in Figure 16, two standardisation groups (3GPP and ETSI) and two industry fora (GSMA and TMFORUM) are taking the lead in building of the edge-cloud in telecommunication business. The following paragraphs provide a brief synopsis of the edge directives set from these initiatives.

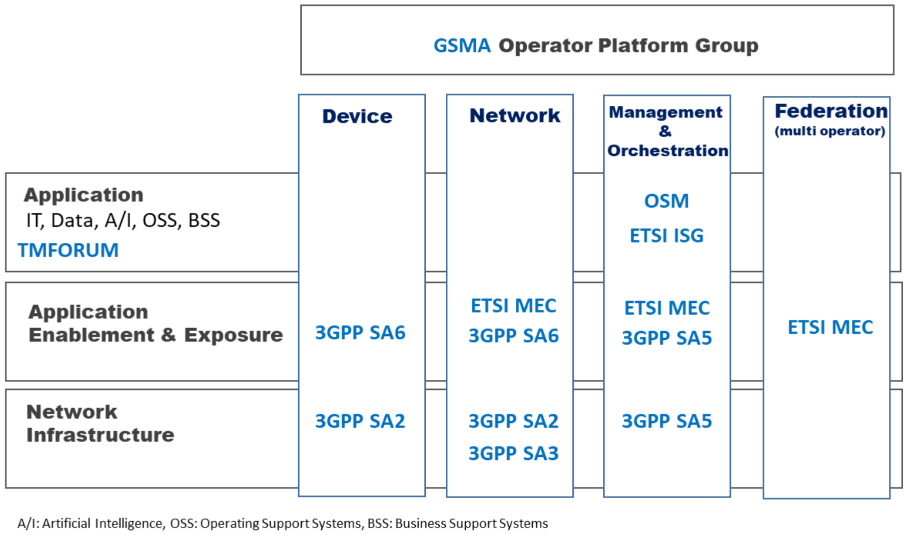

Figure 16 – SDOs & Key Drivers for Edge Computing in Mobile Networks, inspired by Harmonizing standards for edge computing ETSI white paper [1]

4.5.1.1 3GPP for Telecommunications Edge Cloud

The 3rd Generation Partnership Project (3GPP)[2] defines the 3GPP system, covering cellular telecommunications technologies, including radio access, core network and service capabilities. Edge computing in particular has been a major focus in 3GPP Release 17, with four key groups in TSG SA (Technical Specification Group System Aspects) carrying out related studies and normative work [2] as graphically depicted in Figure 17:

[1] https://ocf.etsi.org/research/
[2] https://www.3gpp.org/

- SA2: System Architecture enhancement for supporting Edge Computing.
- SA3: Security aspects for supporting SA2 and SA6 architectures.
- SA5: Management & Charging aspects on Edge Computing.
- SA6: Edge Enabler Layer architecture, and deployment scenarios

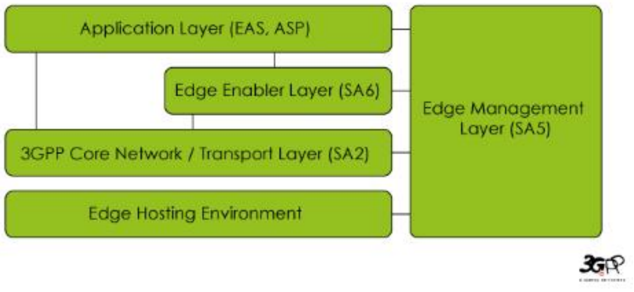

Figure 17 – Simplified View of the 3GPP WGs Edge Work

Fundamental to facilitating edge-cloud deployments is the work led by SA6 on application-enabling service frameworks (also responsible for vertical enablers and mission critical services), that has delivered the following specifications:

- **Common API Framework for 3GPP Northbound APIs (CAPIF)** 3GPP TS 23.222 [3] standardises common capabilities exposed by 5G SA Northbound APIs, for a variety of processes, such as on-boarding/off-boarding Application Functions, service discovery and management, event subscription and notification, security and charging.
- **Service Enabler Architecture Layer for Verticals (SEAL)** 3GPP TS 23.434 [4] specifies a functional architecture to support vertical applications by specifying common application plane and signalling plane entities and services (namely group management, configuration management, location management, identity management, key management and network resource management) that can be shared across vertical applications.
- **Architecture for Enabling Edge Applications (EDGEAPP)** 3GPP TS 23.558 [5] builds upon the concepts set by CAPIF and SEAL and describes the enabling layer and application architecture to implement edge applications on the Edge Data Network (EDN). The enabling layer refers to the exposure of the northbound APIs towards the edge applications, integration with the 3GPP Network and the communication of the application clients running on the UE with the Edge Application Servers (EAS) deployed on the EDN, including capabilities such as service provisioning, rich application discovery and service continuity. Additionally, 3GPP TS 28.538 [6] focuses on Edge Computing Management, addressing Lifecycle management (e.g. on boarding) of Edge applications. GPP TS28.552 [7] and TS 28.554 [8] define the performance measurements and KPIs for edge applications respectively.

4.5.1.2 ETSI MEC for Telecommunications Edge Cloud

The Multi-access Edge Computing (MEC) initiative is an Industry Specification Group (ISG) within ETSI. The work of the MEC initiative aims to unite the telco and IT-cloud worlds, providing IT and cloud-computing capabilities by specifying the elements that are required to enable applications to be hosted in a multi-vendor multi-access edge-computing environment. MEC also enables applications and services to be hosted 'on top' of the mobile network elements, and benefit from being near the customer and receiving local radio-network contextual information. ETSI ISG MEC specified a common and extensible application enablement framework for delivering services, specific service-related APIs for information exposure and programmability, as well as management, orchestration, and mobility related APIs. These APIs facilitate the running of applications at the correct location at the right time and ensure service continuity.

ETSI ISG MEC is currently studying MEC federations to enable shared usage of MEC services and applications across MEC systems in support of a multi-operator/multi-network/multi-vendor environment [1]. ETSI ISG has defined several specifications[3], among which we can highlight the following:

- Multi-access Edge Computing (MEC); Framework and Reference Architecture MEC 003 [9]
- General principles, patterns, and common aspects of MEC Service APIs MEC009 [10]
- Study on Inter-MEC systems and MEC-Cloud systems coordination MEC035 [11]

Since MEC is a design characteristic of the 3GPP 5G Architecture [12][13], ETSI ISG MEC alignment with 3GPP SA2 & SA6 is important and on-going, including aspects related to MEC 5G Integration and future evolution, MEC Federation as well as addressing the operator requirements as set by the GSMA OPG (Operator Platform Group). A very important result of this synergy is the publication of a Harmonised Edge Computing Architecture, as presented in Figure 18, to be used as a blueprint for edge-cloud deployments for the telecom business.

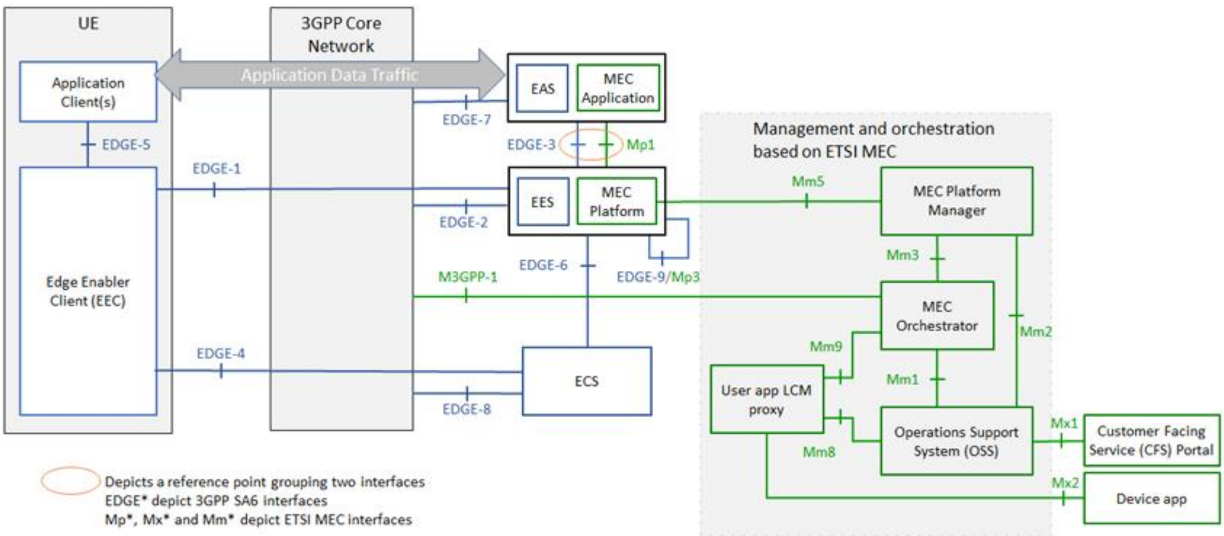

Figure 18 – Harmonized Mobile Edge Cloud Architecture Supported by 3GPP and ETSI ISG MEC specifications [1]

4.5.1.3 GSMA for Telecommunications Edge Cloud

GSMA[4] is a global-led organization within the mobile industry, including 750 mobile operators and over 400 companies in the broader mobile ecosystem, and aims to drive initiatives shaping the future of mobile communications with invaluable insights and industry intelligence in the priority topics of 5G, IoT, Fraud and Security.

[3] https://www.etsi.org/technologies/multi-access-edge-computing
[4] https://www.gsma.com/

Within its Future Networks thread, there are two highlight initiatives related to the edge-cloud momentum for the telecommunications industry, the Operator Platform Group (OPG) and TEC (Telco Edge Cloud) Forum:

- **GSMA OPG & OPAG**: The **Operator Platform Group (OPG)[5]** seeks to support operators to monetize their vast local footprint, their existing relationships with enterprises, and their competence to provide high-reliability services supporting the digital sovereignty through 5G. OPG works on defining a common platform that exposes operator services/capabilities to customers/developers. OPG is open to the wider edge-ecosystem and brings together operators, platform developers, edge cloud providers, Standards Developing Organizations (SDOs), Open-Source Projects, industry 4.0 and market participants. It targets to create the architecture and technical requirements to guide other Standard Developing Organizations (SDOs) in the development of specifications, in the first phase with focus on the Edge, and in future expanding with other capabilities such as slicing. While the Operator Platform Group (OPG) is responsible for the technical requirements and the development of the operator platform, a special subgroup, the **Operator Platform API Group (OPAG)** [14] undertakes the alignment in Operator Platform (OP) APIs fulfilling OP requirements and the collaboration with and contributions to SDOs (such as 3GPP, ETSI and Linux Foundation).

  The OPG project considers the enhancement of the Edge capabilities with:

  - Smart Edge allocation, and selection to perform load deployment and access from the closest edge.
  - Edge federation to offer a multi domain access to customer and enhance edge service under roaming scenarios.
  - Tight network integration to enhance mobility and user experience.
- **TEC Forum**: The Telco Edge Cloud (TEC) Group has a commercial focus bringing together over 20 operators, covering all regions, who are working to promote a collaborative deployment of cloud capabilities at the edge of their networks. TEC is aiming to align Multi Access Edge Computing (MEC business models, charging principles and commercial deployment considerations), and has primary focus on edge cloud trials and POCs. In March 2022, results of early trials were reported in [15].

In summary the Telco Edge Cloud, incorporating the Operator Platform, as graphically depicted in Figure 19, focuses on many edge use cases that are expected to require mobility and sharing of resources and proposes an interoperable (i.e. interface alignment), flexible (i.e. several deployment options) and federated (i.e. one operator can provide access to capabilities managed by another) edge framework.

[5] https://www.gsma.com/futurenetworks/5g-operator-platform/

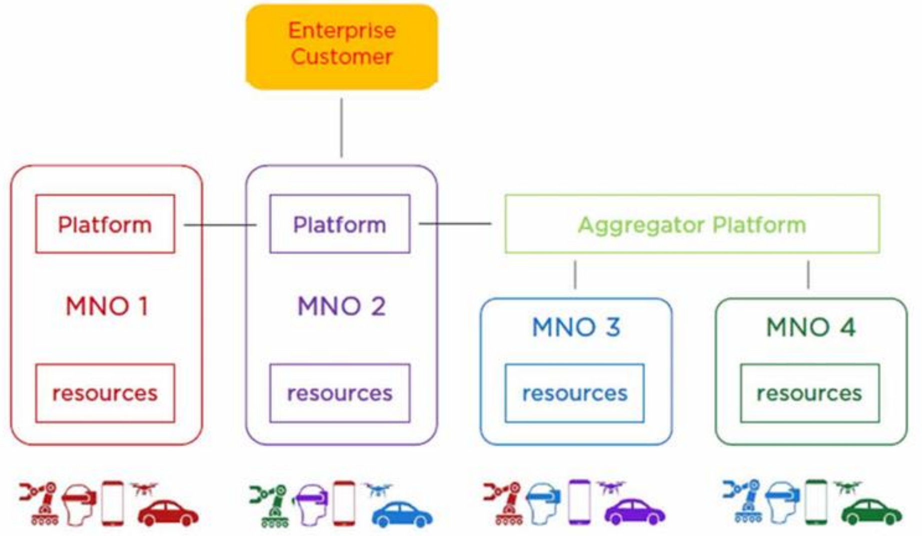

Figure 19 – GSMA Telco Edge leveraging Operator Platform Federation Concept

4.5.1.4 TMForum for Telecommunications Edge Cloud

TMForum is a widely accepted alliance of 850+ global companies with focus of work to help Communication Service Providers (CSPs) towards their digital transformation journey through various ways, from managing the process of transformation through a maturity model, to offering practical toolkits, widely adopted frameworks including Open APIs, as well as, accelerating innovation through rapid POC cases. As of 2020, strategic collaboration of TMForum with GSMA was announced[6], building upon the realization that GSMA's focus on mobile networks and TM Forum's efforts in IT, data and AI make the collaboration between the pair an obvious fit. TMForum is putting effort in both edge computing and Network-as-a-Service (NaaS) and the most relevant work is highlighted below:

- **Open Digital Architecture[7]**: An important initiative towards replacing traditional operations and business support systems (OSS/BSS) with a new approach to building software for the telecoms industry, opening a market for standardized, cloud-native software components, and enabling communication service providers and suppliers to invest in IT for new and differentiated services.
- **Open APIs[8]**: Core to the development of the cloud-native Open Digital Architecture are the OpenAPIs, a suite if application programming interfaces that enable services to be managed end-to-end throughout their lifecycle within an environment where multiple partners might be involved. As an example, API Suite Specification for NaaS[9] supports exposing and managing "Network" Services while more than sixty (60) REST-based Open APIs are developed collaboratively. Up to date, 141 of the world's leading communications

[6] https://www.mobileworldlive.com/featured-content/home-banner/gsma-chair-opens-door-to-tm-forum-collaboration/
[7] https://www.tmforum.org/oda/
[8] https://www.tmforum.org/oda/about-open-apis/
[9]https://www.tmforum.org/resources/specification/tmf909-api-suite-specification-for-naas-v3-0/?_gl=1*sb6rcz*_ga*ODYyNzg4MDUzLjE2NjY3NjcwNzc.*_ga_W21R8NVK4E*MTY2Njg1NDQ5My4zLjEuMTY2Njg1OTYxMC4wLjAuMA..&_ga=2.210490774.2073117244.1666767077-862788053.1666767077

service providers (CSPs) and technology ecosystem participants have signed the Open API Manifesto publicly demonstrating their endorsement of TM Forum's suite of Open APIs.

- **Becoming EDGY** Catalyst Project[10]: The awarded 2019 best new Catalyst in show project sets to explore the maturity of the edge-cloud management solutions in the market to build the ability of dynamic network slicing with zero-touch orchestration, a critical success factor for 5G.
- **The 'EDGE" in Automation** Catalyst Project[11]: Building upon the findings of EDGY, this project demonstrates solutions for Edge Compute as a Service (ECaaS). It aspires to deliver to event developers/suppliers (e.g. concert, sporting/gaming event, exhibition, etc.) pre-scheduled ECaaS packages (e.g. capacity, image recognition, surveillance, doors lock/unlock, emergency services, etc.) at a venue when a crisis occurs, requiring real-time reconfiguration of the edge to deliver public safety emergency services.

Practically as of 2020 and the end of the Catalyst projects, TMForum having endorsed the GSMA OP and TEC specifications has been focusing on the ODA and OpenAPIs work and how to better support Automation at the Edge.

4.5.2 Cloud-native edge architecture

In the ENVELOPE project, we define a common cloud-native edge architecture which includes a set of enabling edge features to support the edge computing requirements as defined in section 3.2. Our approach relies on a containerized architecture design where services are deployed in containers in the different edge hosts. In this way, our architecture abstracts specific implementation choices with respect to the underlying standardization specifications (e.g., ETSI MEC, 3GPP EDGEAPP) and/or open source solutions that will be selected by each trial site, depending on availability of open source edge solutions.

[10]https://www.tmforum.org/collaboration/catalyst-program/becoming-edgy/?_gl=1*4s5ri3*_ga*ODYyNzg4MDUzLjE2NjY3NjcwNzc.*_ga_W21R8NVK4E*MTY2Njg1NDQ5My4zLjEuMTY2Njg1ODI1Mi4wLjAuMA..&_ga=2.242381441.2073117244.1666767077-862788053.1666767077
[11] https://inform.tmforum.org/features-and-opinion/finding-the-edge-in-automation-as-a-service/

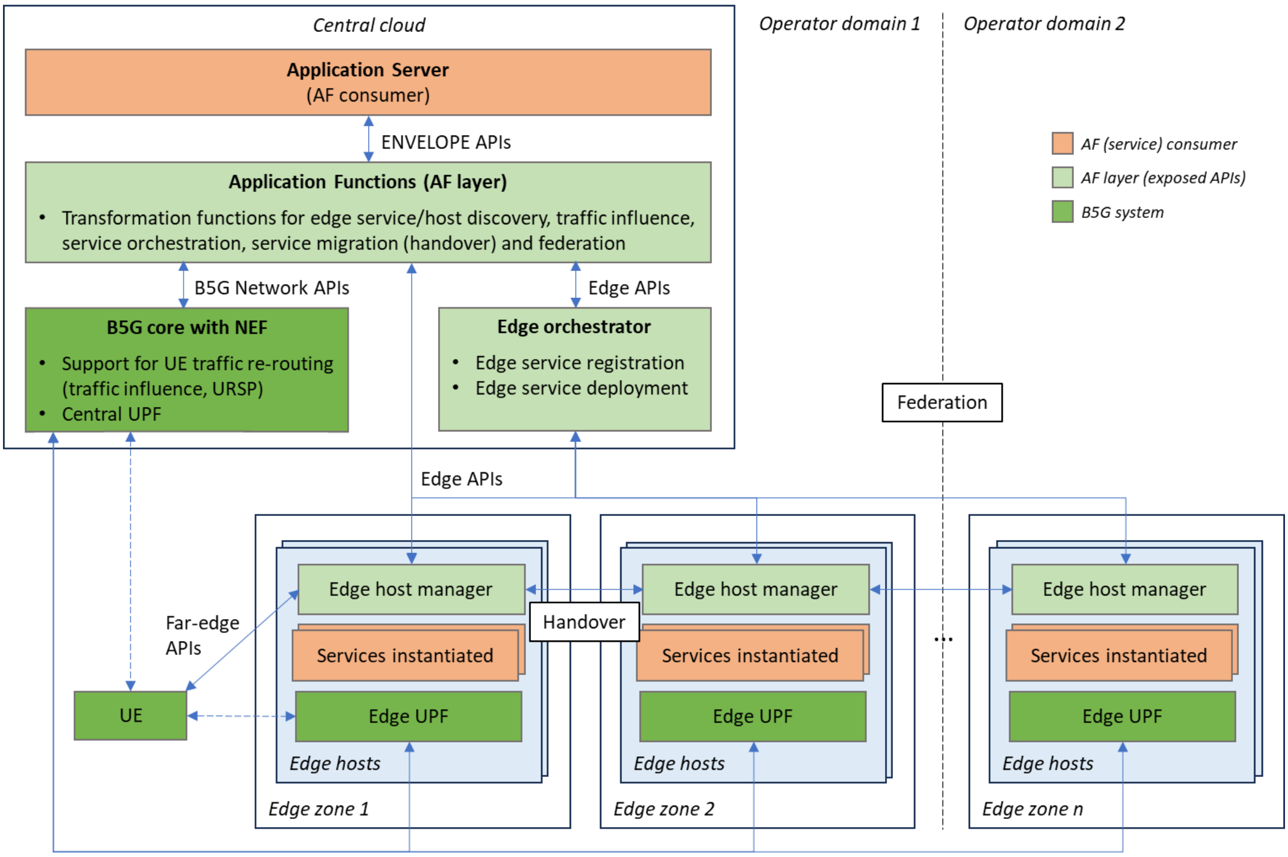

Figure 20 – Common ENVELOPE cloud-native edge architecture

In the ENVELOPE project, we define a common cloud-native edge architecture which includes a set of enabling edge features to support the edge computing requirements as defined in section 3.2. Our approach relies on a containerized architecture design where services are deployed in containers in the different edge hosts. In this way, our architecture abstracts specific implementation choices with respect to the underlying standardization specifications (e.g., ETSI MEC, 3GPP EDGEAPP) and/or open source solutions that will be selected by each trial site, depending on availability of open source edge solutions.

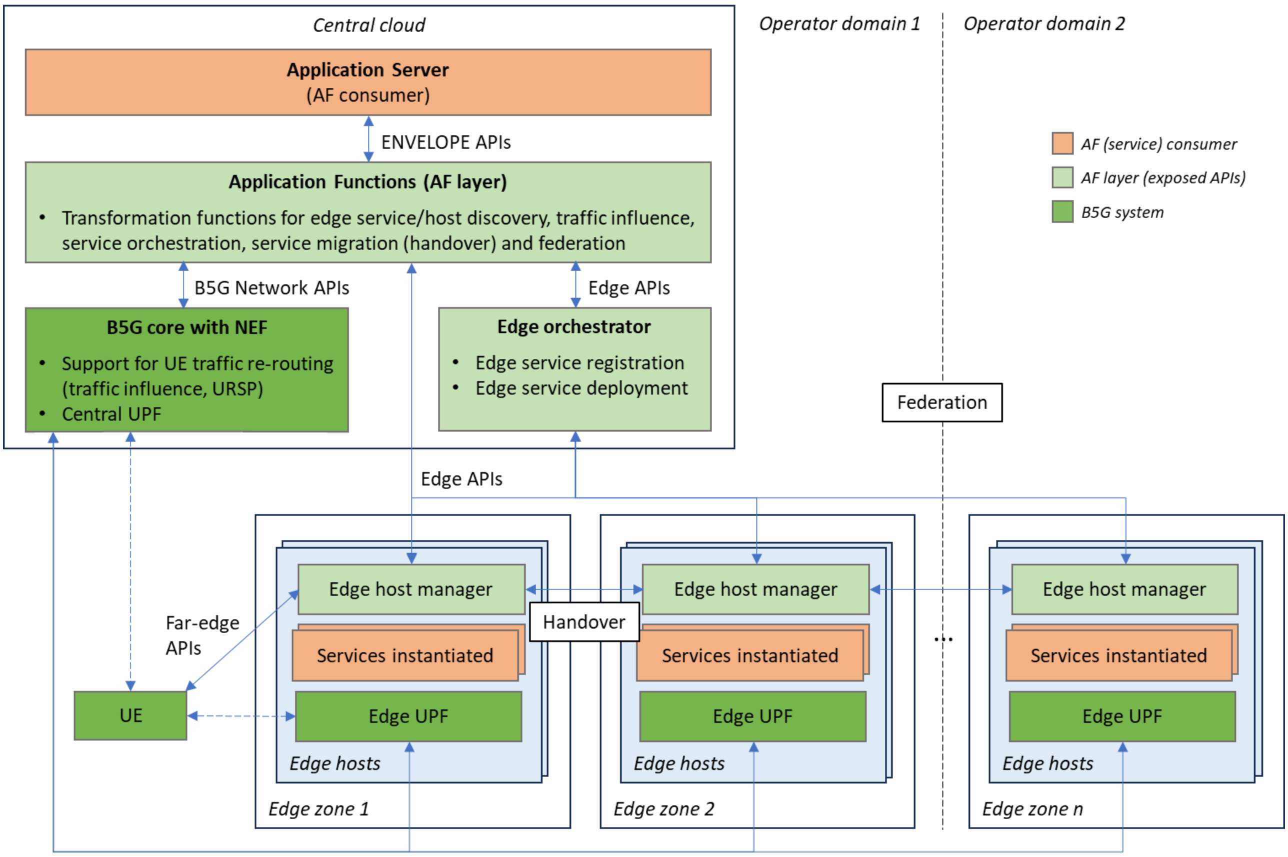

Figure 20shows the common cloud-native edge architecture defined for the ENVELOPE project. Application Functions (AFs) hosted at the central cloud perform transformation functions between the ENVELOPE APIs and edge APIs for underlying B5G and edge functions. These functions include: service/host discovery, traffic influence, edge service orchestration, service migration (handover) and edge federation.

In each telco operator domain, edge hosts are logically grouped into different ‘edge zones’ that could identify, e.g., a group of edge hosts as part of the same physical server infrastructure or as part of the same geographical region within a country. In addition to the services instantiated, each edge host runs a local edge UPF to enable low latency and offloading of data traffic between UE(s) and the selected edge host.

The ‘edge host manager’ performs tasks similar to those specified in ETSI MEC platform or 3GPP Edge Enabler Server (EES) by keeping a list of available services in the host and offering the discovery of these services. Also, service migration (handover) is supported between edge host managers across edge zones via application context relocation.

The edge orchestrator is responsible for the registration and deployment of services in selected edge zone(s) and in the corresponding edge host within the edge zone. This could be done by the Management and Orchestration (MANO) solution available at the server infrastructure.

Federation of edge hosts across domain is supported either via the edge orchestrator or directly between edge host managers when located in different operator domains.

Finally, the AF layer interacts with the B5G core to trigger the UE traffic re-routing from the central UPF towards the edge UPF residing in the selected edge host. This can be done via a new UE route selection policy (URSP) sent to the UE specifying DNNs, slices, and routes to take for a particular application running at the UE. Alternatively, the procedures defined in section 0 for traffic influence can be applied.

4.5.3 ENVELOPE edge APIs

With respect to ENVELOPE APIs, CAMARA[12] defines a few "Edge Cloud" service APIs that will be used in the project. CAMARA is an open-source project launched at Linux Foundation that defines, develops, and publishes GSMA Open Gateway northbound service APIs. The CAMARA edge cloud APIs enable service providers to:

- Simple Edge Discovery / Application Endpoint Discovery APIs discovers the closest edge cloud zone with respect to a given device, where 'closest' could mean, e.g., number of hops between the device and the edge zone.
- Application Management API provides and manages compute resources within the operator network to deploy application images on VMs or containers.
- Traffic Influence API influences the traffic routing from the user device towards the edge instance of the application.

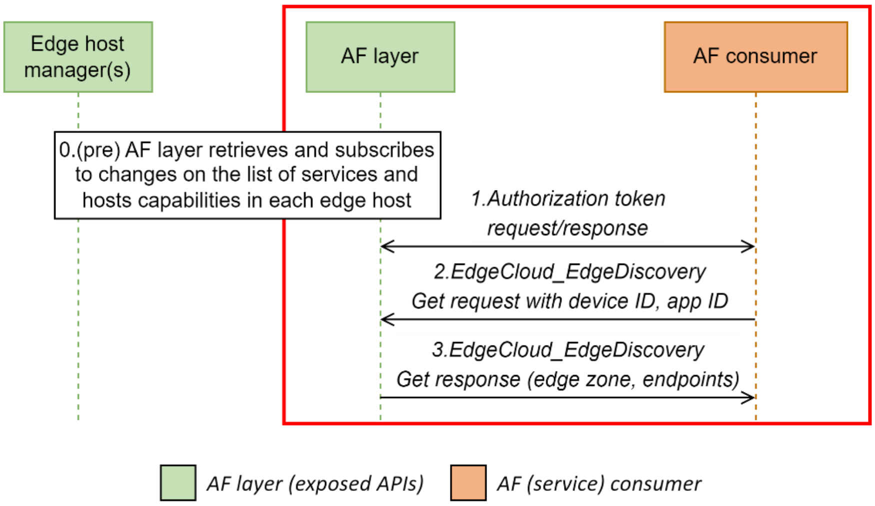

Figure 21 – Possible message flow for edge discovery with CAMARA edge cloud APIs

Figure 21 shows one possible message flow for the discovery of edge zones when using the CAMARA edge cloud API with the following steps:

0. As pre-step, the AF layer retrieves and subscribes to changes on the list of services and host capabilities in each edge host of all available edge zones.
1. The AF consumer performs authentication with the AF layer to be able to consume the corresponding edge cloud API for edge zone discovery. The authentication is done via security access keys such as OAuth 2.0 client credentials.
2. The AF consumer sends a request to discover the best (or a list of) edge zone(s) for a certain device given the device ID and optionally the application ID intended to be reached.

[12] https://camaraproject.org/

3. The AF layer replies with the list of edge zones and optionally the application endpoint residing in the edge host selected within the edge zone.

At the moment, the Simple Edge Discovery and Application Endpoint Discovery APIs from CAMARA only support the filter (criteria) 'closest' to the device. Therefore, extensions are needed to support additional filters such as 'edge host capabilities' if specific computing resources are required (e.g., GPU).

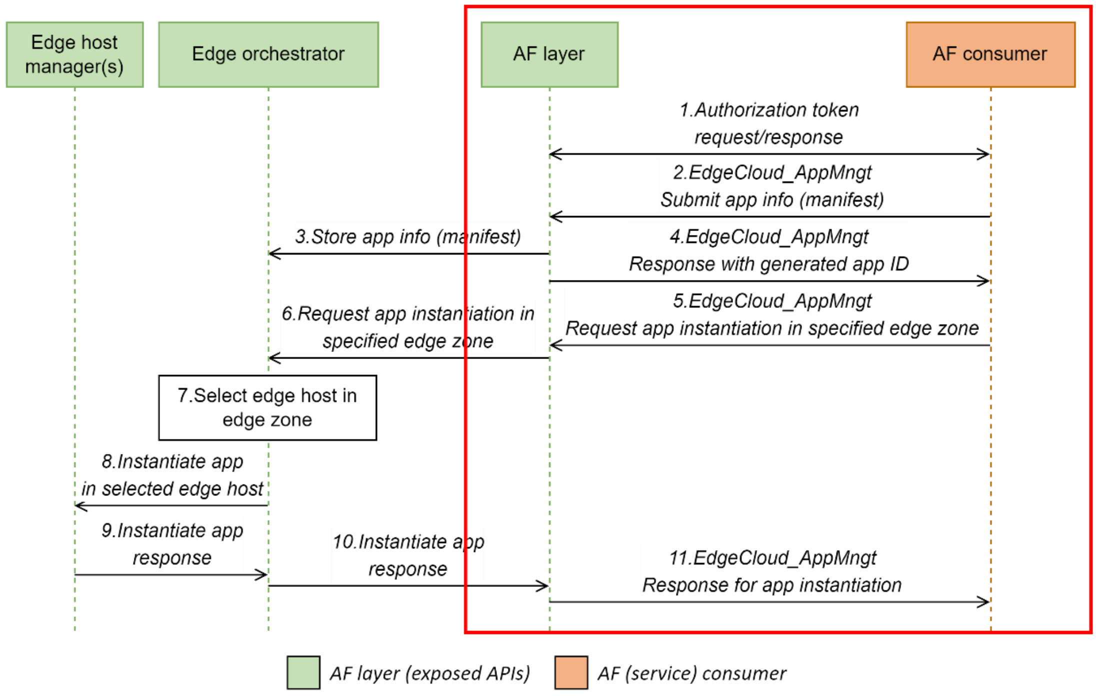

Figure 22 – Possible message flow for edge application management with CAMARA edge cloud APIs

One possible use of the CAMARA edge cloud Application Management API is shown in Figure 22 with the steps:

1. The AF consumer performs authentication with the AF layer to be able to consume the corresponding edge cloud API for application management. The authentication is done via security access keys such as OAuth 2.0 client credentials.
2. The AF consumer registers the application at the AF layer by sending manifest info with metadata about the application to be registered.
3. The AF layer converts this metadata to corresponding format as supported by the edge orchestrator.
4. The AF layer replies with a confirmation of the application registration to the AF consumer. An application ID is generated and given to the AF consumer.
5. When the AF consumer decides to instantiate an application previously registered, it sends a request to the AF layer specifying at which edge zone the application must be instantiated.
6. The application instantiation is requested to the edge orchestrator.
7. The edge orchestrator selects an available and suitable edge host within the chosen edge zone for instantiating the application.
8. The edge orchestrator triggers the instantiation of the application at the selected edge host.
9. An instantiation confirmation is sent from the edge host to the edge orchestrator.

10. An instantiation confirmation is sent from the edge orchestrator to the AF layer.
11. Finally, the AF consumer receives a final confirmation that the selected application has been instantiated.

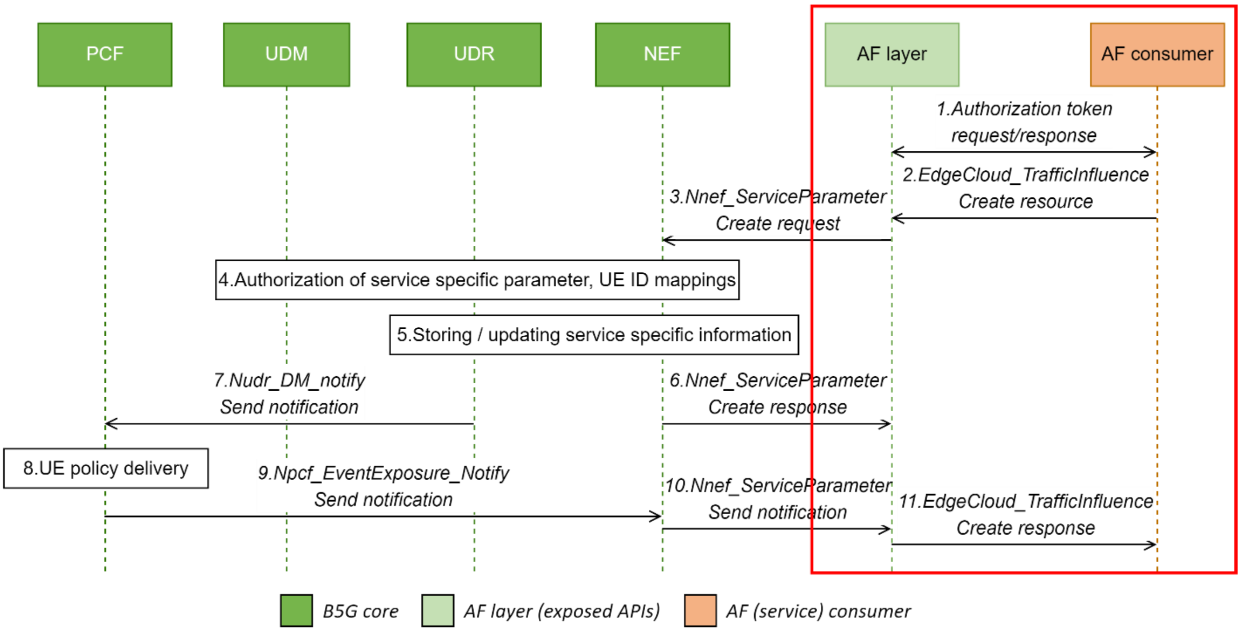

Figure 23 – Possible message flow for edge traffic influence with CAMARA edge cloud APIs

Finally, Figure 23 shows one possible message flow for the traffic influence when using the CAMARA edge cloud API with the steps as defined by 3GPP specification TS 23.502, section 4.15.6.7 [13]on service specific parameter provisioning:

1. The AF consumer performs authentication with the AF layer to be able to consume the corresponding edge cloud API for traffic influence. The authentication is done via security access keys such as OAuth 2.0 client credentials.
2. The AF consumer request traffic influence for the data traffic of a given device to be directed towards an application residing at a given edge zone.
3. The AF layer requests the NEF to update service parameters for a particular service (e.g., allowed DNNs, S-NSSAIs).
4. The NEF may need to authorize the service specific parameter provisioning request with the UDM. Also, the NEF may perform mappings between external UE IDs and internal IDs.
5. The NEF stores the AF layer request information in the UDR as the "Application Data".
6. A service parameters update confirmation is sent from NEF to the AF layer.
7. The PCF is notified about service specific changes at the UDR.
8. The PCF initiates the delivery of the new/updated UE Policy.
9. The NEF receives a confirmation that the UE policy has been sent to the device.
10. The NEF confirms the service parameter update to the AF layer.
11. Finally, the AF layer sends a confirmation for the traffic influence to the AF consumer.

[13] 3GPP, "5G; Procedures for the 5G System (5GS)," 3GPP TS 23.502, Rel. 18, May 2024

4.6 Enablers for ATSSS-like multi-connectivity

4.6.1 State of the Art

Access Traffic Steering-Switching-Splitting (ATSSS) is a relatively new approach introduced in the 3GPP 5G Release 16 to provide redundancy and reliability within the network, by allowing the traffic flow to utilize both the 3GPP compliant 5G path, as well as the non-3GPP compliant path (usually WiFi) [19, 20]. ATSSS allows the traffic to be steered, switched or split between the available multiple paths, based on their availability and the network conditions. **Steering** of a traffic flow refers to the initial selection of an access path to carry the data of the traffic flow, switching is the **seamless** transition of a traffic flow from one access path to the other, and **splitting** is the simultaneous use of both access paths for transmitted the data of a traffic flow.

A simplified ATSSS architecture based on the 3GPP technical specifications [16] is illustrated in the figure below. In this setup, the UE registers to the common 5G core network via both 3GPP access and non-3GPP access. In addition, the UE establishes a so-called Multi-Access (MA) PDU Session that creates a 3GPP access connection and a non-3GPP access connection between the UE and a common UPF anchor, which is selected by the control plane during the MA PDU Session establishment. Based on network policy, the control plane generates traffic steering rules and distributes them to UE and UPF anchor. These rules dictate how uplink and downlink traffic should be distributed across the two access connections and enable dynamic resource utilization and seamless service continuity.

The UE is equipped with multiaccess steering functionalities such as Multipath TCP (MPTCP), Multipath QUIC (MPQUIC), and ATSSS-Low Layer (ATSSS-LL) [16] to enable advanced traffic steering, switching, and splitting capabilities for various types of traffic, e.g., UDP, TCP, IP, Ethernet. These components in the UE collaborate with the corresponding multiaccess proxy functionalities in the UPF anchor to ensure efficient multiaccess connectivity in the uplink (UL) and downlink (DL) directions across the two access connections based on the corresponding traffic steering rules provided by the control plane.

The role of the Performance Measurement Function (PMF) [16] within both the UE and UPF anchor is to dynamically measure performance metrics on each access connection (including RTT and Packet Loss Rate) and adjust the traffic routing accordingly. By measuring these metrics, it is feasible to route traffic to the access with the smallest delay or to the access with the smallest packet loss rate.

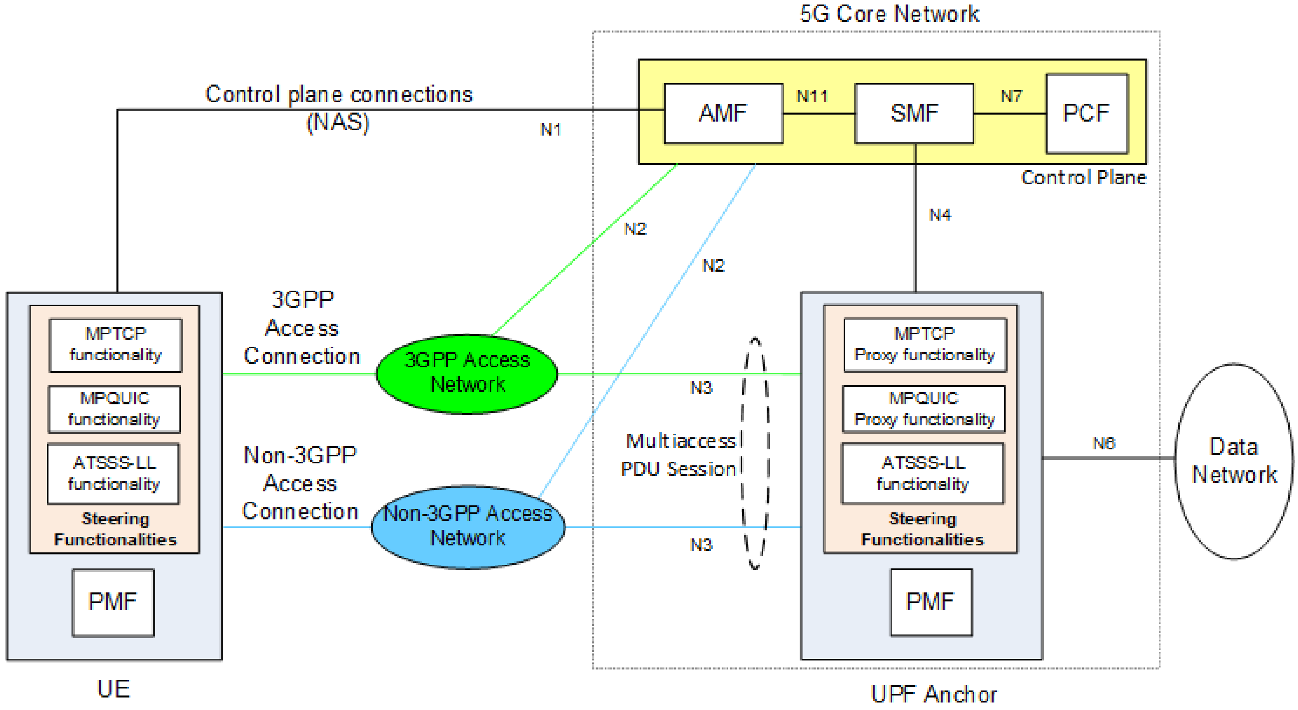

Figure 24 – Simplified ATSSS architecture based on 3GPP specifications.

The UE and UPF distribute data traffic across the two access connections on the uplink and downlink direction respectively, using traffic steering rules provided by SMF. The traffic steering rules – called ATSSS rules in the UE and N4 Multi-access rules in UPF – are generated based on multiaccess control policy configured in the PCF (e.g., by OAM).

An example of an ATSSS rule provided to the UE is shown below:

- Rule identity
- Rule precedence
- Traffic descriptor
  - Application identity: com.example.app1
  - Protocol: TCP
- Access Selection descriptor
  - Steering functionality: MPTCP
  - Steering mode: Load-balancing
  - Steering Mode Information: 20% over 3GPP and 80% over Non-3GPP
  - Threshold Values: Max Round-Trip-Time (RTT) = 20ms

This ATSSS rule specifies that the TCP traffic of application "com.example.app1" should be steered by the MPTCP steering functionality using a load-balancing steering mode that must send 20% of the traffic over 3GPP access and 80% of the traffic over non-3GPP access, provided that the RTT of each of these accesses does not exceed 20ms. If the RTT of an access exceeds 20ms, then no traffic should be sent on this access.

Despite advancements in 3GPP standards, a notable limitation persists in the current open-source programmable software stacks for 5G technology: the absence of key multi-access and ATSSS feature implementations. Open-source solutions such as Open5GS [21] and Free5GC [22], while extensively utilized within the research community, they do not presently support the functionality required for conducting research on 5G multi-access networks and ATSSS [23].

4.6.2 ATSSS architecture

To work around the above limitations of the open source 5G implementations (i.e., the lack of native ATSSS support), ENVELOPE adopts an over-the-top approach for supporting multi connectivity. In general, ENVELOPE will support an ATSSS enabler/configuration, which when triggered by the vertical application activates multipath connectivity based on ATSSS rules.

The ATSSS-like multi-connectivity architecture that will be supported in the project is shown in the figure below. The overall design is optimized for low-latency and high-reliability scenarios, crucial for CAM services. It allows seamless traffic handover and redundancy across Wi-Fi and 5G, ensuring service continuity.

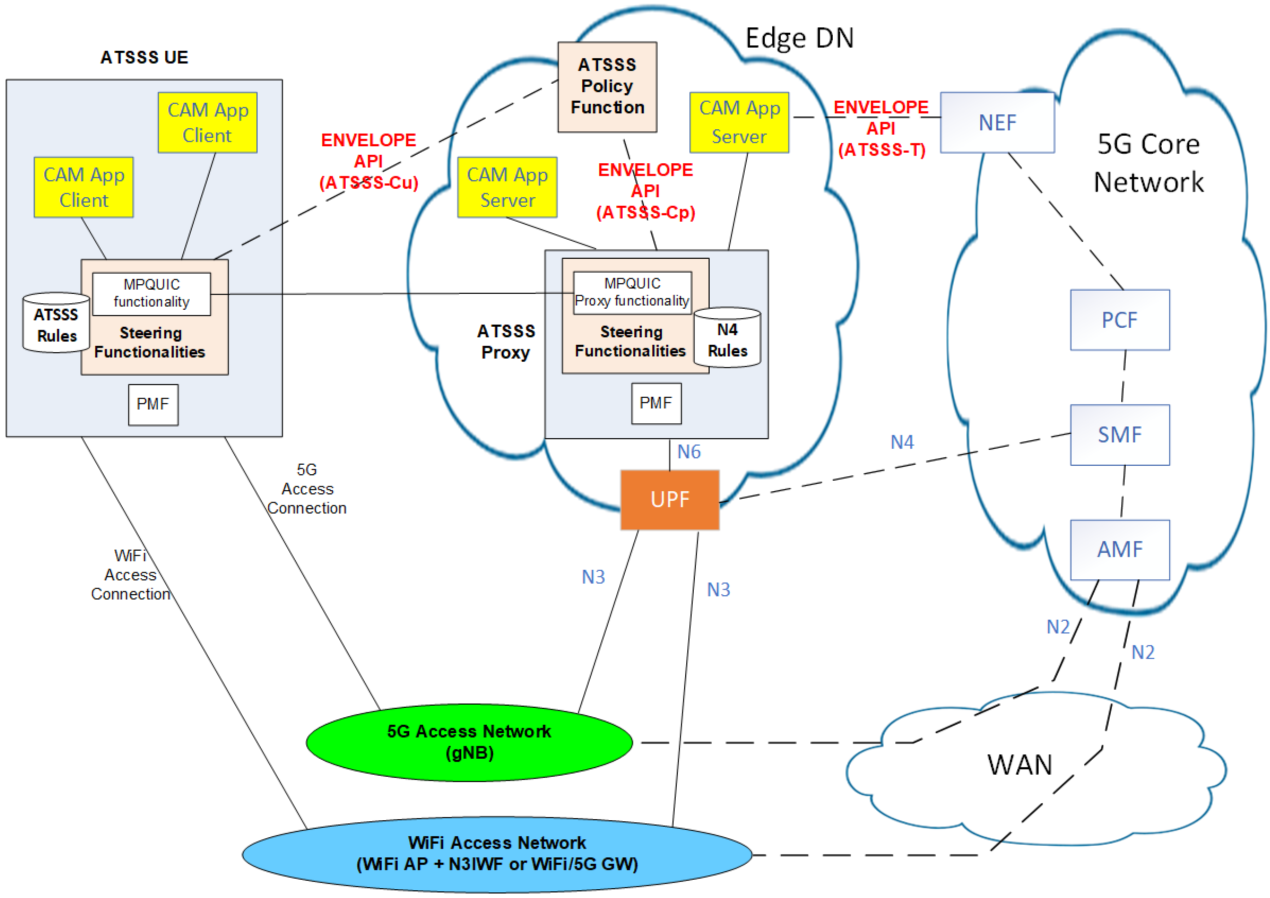

Figure 25 – Multi-connectivity architecture in ENVELOPE.

On the network side, the architecture introduces a novel function called "ATSSS proxy", deployed in an edge data network. The CAM applications in the UE communicate with the corresponding CAM applications in the edge network by using multi-access communication via the ATSSS proxy.

The UE and the ATSSS proxy implement an MPQUIC-based steering functionality based on the Multipath QUIC (MPQUIC) protocol defined in IETF draft and the 3GPP specifications in TS 24.193. Details are provided in the following section.

The ATSSS Policy Function provides ATSSS rules to UE, specifying how uplink traffic should be allocated between the two access connections. Similarly, it supplies ATSSS Proxy rules to the ATSSS Proxy, dictating how downlink traffic should be distributed across the access connections. This allows for adaptive and optimized traffic handling based on real-time network conditions and application requirements. Although the ATSSS Policy Function is shown within an edge data network, it could also be deployed in a centralized data network.

The ENVELOPE APIs shown in the figure are discussed further below in clause 4.6.3.

4.6.3 MPQUIC Steering functionality

By default, the traffic will only flow over a single defined 5G path. Once the vertical service activates the multi-connectivity option through the ENVELOPE interface then ATSSS signalling starts. Note that due to the required timescales the vertical application does not influence the specific ATSSS policy function. However, the vertical application may install different intelligent rules (e.g. trained Neural networks) that guide this decision process on the Learning Agent sides at both the UE and the edge DN. A learning agent in the context of a trained neural network is an AI system designed to adapt, improve, and make better decisions over time by interacting with its environment. Unlike traditional AI systems that operate based on fixed rules, learning agents evolve through experience

Once ATSSS is triggered by the vertical service the ATSSS proxy in the edge DN and the UE initiate their MPQUIC functionalities to enable simultaneous use of both paths, according to the IETF draft [25]. The WiFi path is activated between the UE and DN with a unique path_ID and associated connection_ID. Both paths (5G and WiFi) will also maintain distinct packet number states for sending and receiving packets, to enable path-based packet recovery and congestion control mechanisms, as defined in IETF draft [25]. The realization of the MPQUIC proxy functionalities will be based on an extension of the picoQUIC library [24] such that it creates individual paths and maps flows of traffic to each available path. No changes to the data plane, specifically the UPF are required in this approach.

In our proposed approach, the ATSSS policy function controls the MPQUIC steering functionality and the mapping of flows of traffic to each available path, by deciding which ATSSS rule to implement next, based on the network conditions and application requirements. The ATSSS policy function is learned using reinforcement learning which takes as input the real-time network conditions and application requirements and outputs the optimal ATSSS rule for the next T seconds. This rule is then used both for uplink and downlink traffic. The set of available ATSSS rules as per the 3GPP technical specifications [19] are: (i) active-standby, (ii) smallest delay, (iii) load-balancing, (iv) priority-based, and (v) duplication. Additionally, if the UE receives from the ATSSS policy function load-balancing as the rule, it uses the reinforcement learning based ATSSS rules available within the UE to decide the splitting ratio for the load-balancing rule. This splitting ratio is then also sent to the ATSSS proxy in the Edge DN for the downlink traffic management.

ENVELOPE ATSSS APIs

The ENVELOPE platform shall support the following APIs for the ATSSS-like multi-connectivity enabler.

API name	Description	Modules Involved	Use Cases
ATSSS-T	Enables a vertical service (e.g., a CAM App Server) to request via NEF multi-connectivity service for its data traffic.	NEF, AF	GR-UC-1
ATSSS-Cu	Enables an ATSSS UE to retrieve multi-connectivity traffic steering rules from an ATSSS Policy Function.	UE, ATSSS Policy Function	GR-UC-1
ATSSS-Cp	Enables an ATSSS Proxy to retrieve multi-connectivity traffic steering rules from an ATSSS Policy Function.	ATSSS Proxy, ATSSS Policy Function	GR-UC-1

4.7 Enabler for Predictive QoS

4.7.1 State of the Art

Predictive QoS (PQoS) has long been considered as a promising approach in enabling the graceful adaptation of vertical services/application to the anticipated conditions in the network. Applying analysis on historical and contextual data that describe network performance and conditions, PQoS enables the proactive identification of significant fluctuations and the generation of corresponding notifications (i.e., In-advance Quality of Service Notifications, IQNs) towards the vertical service/application, allowing the later to adjust its operation within the anticipated environment[14]. Currently, as per 3GPP specifications[15] the NWDAF (Network Data Analytics Function) supports the exposure of analytics, including reports on either historical data, or predictions/estimated values of selected metrics. The NWDAF can contain an Analytics logical function (AnLF), for the generation of the analytics information, and/or a Model Training logical function (MTLF) that involves the training process of a ML model, practically enabling the preparation of a AnLF instance.

The exposure of analytics is supported in two different modes, namely: (i) a subscription/notification mode (Nnwdaf_AnalyticsSubscription) where the delivery of analytics is triggered by the NWDAF subject to certain conditions e.g., threshold KPI value reached; and (ii) a request/response mode (Nnwdaf_AnalyticsInfo_Request) where the analytics information is provided on demand by the consuming function. In both cases a series of different types of analytics are foreseen e.g., Slice load level information; Service experience; QoS sustainability; DN performance; etc. 3GPP TS 23.288 [26] Clause 6 specifies the procedures and interfaces utilized for the retrieval of the

[14] https://5gaa.org/content/uploads/2023/01/5gaa-wi-presa-tr-predictive-qos-and-v2x-service-adaptation.pdf
[15] TS 23.288

necessary data from the various entities/components/nodes of the 5G system i.e., RAN, 5G Core network functions, and beyond e.g., Application Server/Function (see also 3GPP TS 23.436[16]).

While Predictive QoS has attracted significant attention throughout the last years, there is no 3GPP-compliant implementation available, to the best of our knowledge. A very limited number of 3GPP compliant implementations of NWDAF have been released during the last years [33], [34], [35] however they are either not currently actively maintained[17] or they are tailored for 5GC implementations other than the ones currently considered by the project[18].

4.7.2 Predictive QoS architecture

The ENVELOPE platform aims to provide applications/services with access to three types of **PQoS Inferense (PQoS-I) services**. These services shall provide vertical applications with notifications regarding predicted fluctuations of the upcoming quality of service. The services differentiate on whether the take into account application-level aspects as explained below. In all cases we target a subscription/notification model, so as to generate notifications only when needed i.e., expecting to cross a threshold.

QoS Sustainability PQoS (QS-PQoS) service. Here predictions refer to network level performance metrics. The targeted functionality fits the scope of the QoS Sustainability type of analytics specified in 3GPP TS 23.288 Clause 6.9, where the notifications express “the likelihood of a QoS change for an Analytics target period in the future in a certain area”. The input data include metrics attained by the OAM (including RAN metrics), AMF, SMF.

DN Performance PQoS (DNP-PQoS) service. Here predictions refer to service level (end-to-end, e2e) performance metrics. The targeted functionality fits the scope of the DN Performance type of analytics specified in 3GPP TS 23.288 Clause 6.14, where the notifications capture “user plane performance (i.e. average/maximum traffic rate, average/maximum packet delay, average packet loss rate)”. According to the 3GPP specifications the main source of input data in this case is the AF.

Hybrid PQoS (H-PQoS) service. Here predictions refer to service level performance metrics, however prediction considers both input data from the network (QS-PQoS input data) and service level input data (DNP-PQoS input data).

The first phase of the project (i.e., up to Open Call 1) will focus on the delivery of the aforementioned services, building on ML models pre-trained via offline procedures. The second phase of the project shall focus on the delivery of **PQoS Training (PQoS-T) services:** these services shall provide

[16] TS 23.436 (TODO: Expand)

[17] A more detailed analysis of the currently available implementations will be provided in the context of WP3.

[18] A. A. Bayleyegn, Z. Fernández and F. Granelli, "Real-Time Monitoring of 5G Networks: An NWDAF and ML Based KPI Prediction," *2024 IEEE 10th International Conference on Network Softwarization (NetSoft)*, Saint Louis, MO, USA, 2024, pp. 31-36, doi: 10.1109/NetSoft60951.2024.10588931.

vertical applications with the ability to train their individual ML models, tailored for the support of individualized DNP/H-PQoS services.

The following figure illustrates the high level architecture of the ENVELOPE PQoS services solution. The AF Consumer represents the vertical service consuming the PQoS services. This is performed through a simplified ENVELOPE API which abstracts the details of the 3GPP TS 29.520 API details, as described in the following subsection. The AF Layer is a transformation function of the ENVELOPE Platform responsible for the translation between the ENVELOPE API and the underlying 3GPP interface towards NWDAF. The latter contains a Controller entry component responsible for handling incoming requests. These may be EventSubscriptions, AnalyticsInfo Requests or requests for PQoS-T. An AnLF instance corresponds to the ML model responsible for generating the predictive analytics. Predictions are based on periodic inference calls to the AnLF. All input data for generating the analytics is retrieved from a Monitoring DB. For event subscriptions, a periodic process retrieves the latest input data, performs an inference call and checks the Reporting Thresholds based on which it generates or suppresses a notification (see also next). The Monitoring DB further also feeds the training of MTLF instances by providing historical data. All monitoring data are asynchronously retrieved by the Monitoring DB from various locations. Namely:

- **UE**: application layer KPIs e.g., Round Trip Time (RTT), DL/UL data rates, location information (latitude, longitude, elevation)
- **gNb**: aggregate data rate, radio conditions (RSRP, RSRQ, RSSI, SNR, CQI)
- **AF**: application layer KPIs
- **EAS** (Edge Application Server): resource utilization (CPU/GPU, RAM, I/O)
- **5GC**: number of connected UEs at gNb

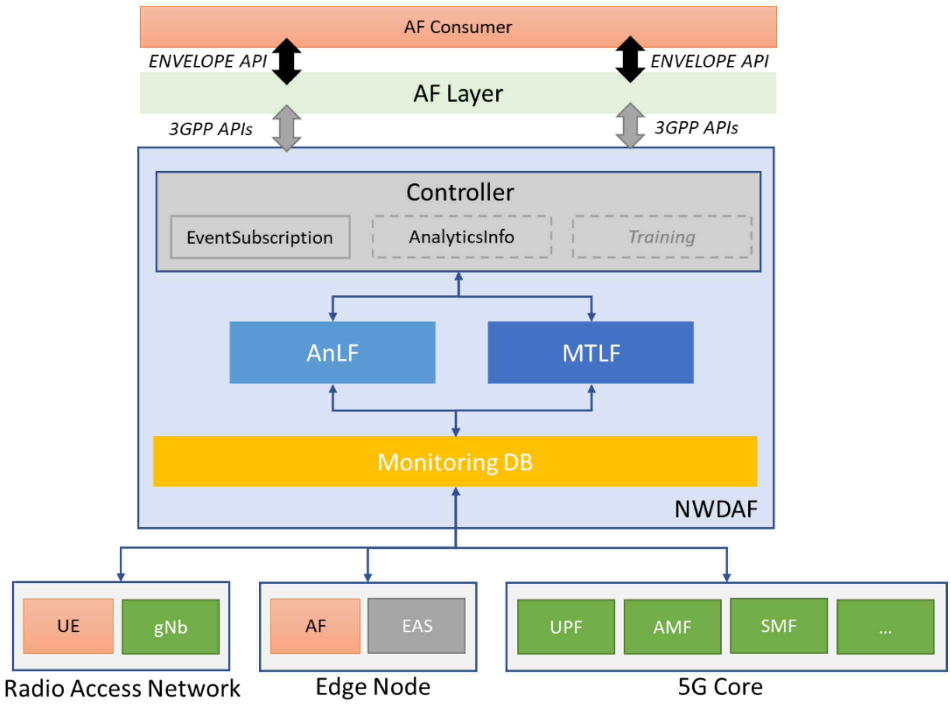

Figure 26 – High-level pQoS architecture

4.7.3 ENVELOPE Predictive QoS APIs

The ENVELOPE platform shall support two APIs, namely for PQoS Inference and Training.

API name	Description	Modules Involved	Use Cases
PQoS-I	Enables vertical service to subscribe to notifications on network level or service level QoS KPI changes predicted.	NWDAF	GR-UC-1
PQoS-T	Enables vertical service to configure the ENVELOPE platform to train a DNP/H-PQoS ML model.	NWDAF	GR-UC-1

The following figure presents the signalling sequence for the provision of the ENVELOPE PQoS-I services.

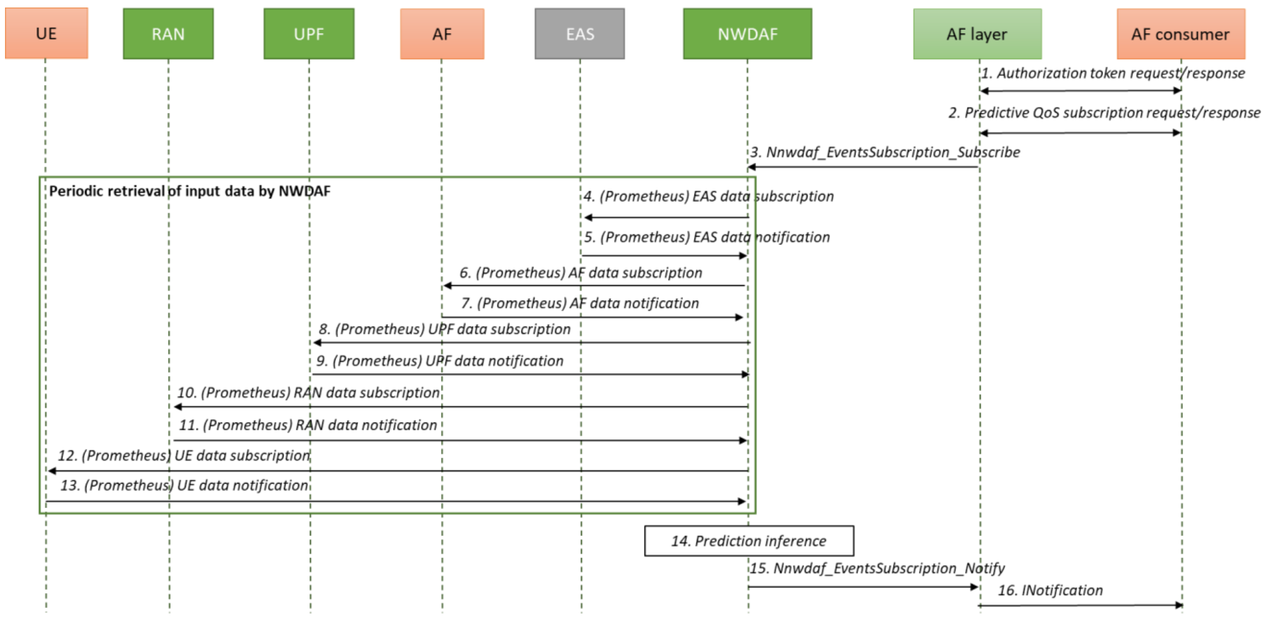

Figure 27 – PQoS-I service sequence diagram

1. The AF consumer performs authentication with the AF layer to be able to consume the corresponding API. The authentication is done via security access keys such as OAuth 2.0 client credentials.
2. The AF consumer submits a subscription request to the NWDAF. The request includes information regarding:
    a. PQoS service type i.e., Qos Sustainabiliity, DN Performance or Hybrid,
    b. a target performance metric (analytics target), its reporting threshold value i.e., the estimated value triggering a notification, and the time horizon of the prediction.
    c. the traffic flow direction (UL/DL) if applicable.
3. The AF layer translates the AF consumer subscription information into a Nwdaf_EventsSubscription Subscribe message. This interaction complies with the Nnwdaf_EventsSubscription Service API as defined in TS 29.520 Clause 5.1.

NOTE A: Step 3 triggers the periodic retrieval of input data by the NWDAF, from various entities in the overall system as described in the previous section. The exact selection of entities and corresponding input data depends on the selected service (QS/DNP/H) and user input in step 2.

NOTE B: For simplification reasons the sequence diagram does not include the retrieval of the ML model for the support of the service. Here it is assumed that a suitable model exists and the subsequent steps focus on retrieving the input data for the inference step. Figure Y further below focuses on the retrieval of the model, including the support of the PQoS-T service.

4. The NWDAF subscribes to EAS data via a non-3GPP API which relies on Prometheus exporter. Applicable in DNP/H-PQoS only.
5. The EAS periodically provides the requested data.
6. The NWDAF subscribes to AF data via a non-3GPP API which relies on Prometheus exporter. Applicable in DNP/H-PQoS only.
7. The AF periodically provides the requested data.
8. The NWDAF subscribes to UPF data via a non-3GPP API which relies on Prometheus exporter.
9. The UPF periodically provides the requested data.
10. The NWDAF subscribes to RAN data via a non-3GPP API which relies on Prometheus exporter.
11. The RAN periodically provides the requested data.
12. The NWDAF subscribes to UE data via a non-3GPP API which relies on Prometheus exporter. Applicable in DNP/H-PQoS only.
13. The UE periodically provides the requested data.
14. The NWDAF produces a prediction regarding the requested metrics (as defined in step 2).
15. When a notification is generated, a Nnwdaf_EventsSubscription_Notify message is created by the NWDAF and sent to the AF layer.
16. The AF layer delivers the notification to the AF consumer.

Step 2 in the above sequence aims to simplify the interaction of the vertical service with the NWDAF API, which provides an all-inclusive but also rather complex set of configuration options. In the following list we provide examples of Nnwdaf_EventsSubscription_Subscription[19] information that is abstracted by ENVELOPE, in particular for the Analytics Filter Information:

- QoS requirements: translating KPIs indicated by the AF consumer to 5QI identifiers.
- Location information: a default Area Of Interest is selected by the platform based on the Cell ID information of the trial site.
- Anchor UPF info: the correct value shall be configured by the platform itself rather than the AF consumer.
- Application Server Addresses: the correct value shall be configured by the platform itself rather than the AF consumer.
- Etc.

Figure 28 below presents the sequence diagram for the PQoS-T service.

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[19] 3GPP TS 29.520, Clause 5.1.6

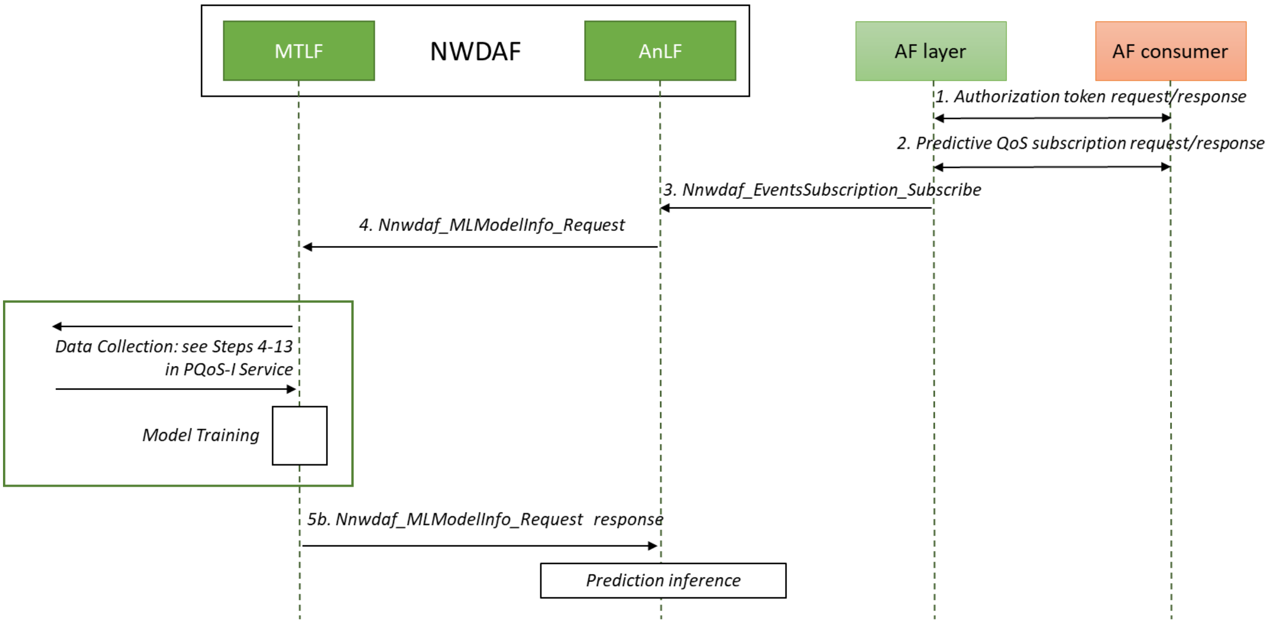

Figure 28 – PQoS-T sequence diagram

1-3. The first three steps are the same with the PQoS-I service. This is because the PQoS-T service gets to be provisioned in the case that there is no ML available to support the requested PQoS-I service.

4. This step takes place when the NWDAF (AnLF) does not find any suitable ML model for the provision of predictive analytics. In this step the AnLF instance places a Nnwdaf_MLModelInfo_Request towards the MLTF following the specifications of 3GPP TS 23.288 Clause 6.2A.
5. (a) If the MLTF finds no fitting ML model it triggers the collection of the required monitoring information as described in Steps 4-13 of the PQoS-I sequence diagram. The identification of the desired metrics is based on the AF subscription message in step 2.
   (b)If the MLTF finds a fitting ML model (or a newly spawned training process has completed), it responds to the AnLF with a Nnwdaf_MLModelInfo_Request response message with all necessary information for retrieving and using the trained model.

4.8 Roaming

International mobile roaming is a service that allows mobile users to use their mobile devices for voice calls text messages and data calls while visiting another country. 3GPP has provided detailed specifications for the roaming architecture, clarifying the control and user data flow between the Home (HPLMN) and the Visited (VPLN) network. Practical implementations are streamlined through a concrete set of roaming guidelines extended to support charging and security aspects.

In CAM services, roaming is very important, considering that vehicles are produced in one country, sold in another, owned in a 3rd and can be driven across numerous countries and continents. Car manufacturers on-boarding SIMs at the production plants face roaming concerns both in terms of subscription management as well as roaming charging fees from the start. OEMs of OBUs and modems face the challenge of universally supporting all possible frequencies as regulated per region/country.

Two operational modes of roaming are foreseen, the Home-Routed (HR) where the user traffic is always served by the home network (HPLMN) and the Local Break Out (LBO) where the user's traffic is served locally by the visited network (VPLMN). HR is the current practice and considering that roaming networks get interconnected through public GRX/IPX hubs optimized for one-to-many connections and not performance, it is highly unlikely that the CAM requirements on end-to-end latency can be supported. The most relevant mode for CAM use cases is the LBO, which complements well with the performance gains of processing at the edge.

Focusing on the national cross-borders, roaming becomes increasingly important for the CAM use cases, as service continuity during the hand-over among networks (inter-PLMN handover) must be maintained and the requested service KPIs must be guaranteed. It is noteworthy that current roaming solution mandates that while moving from the coverage of HPLMN to VPLMN all active sessions will be terminated, and a network reselection process will be initiated to properly select the target VPLMN, a process that can take from few seconds to minutes. On top, the existing roaming procedures have not been designed for differentiated services and are not yet addressing the fundamental CAM requirements on traffic prioritization and guaranteed QoS pending the adoption of the latest 3GPP developments. 3GPP 23.501 is the baseline to address CAM challenges, addressing both the appropriate roaming LBO implementation as well as in respect to service differentiation through slice roaming specifications. Putting a step further, GSMA has already provided the roaming guidelines for the 5G SA roaming, endorsing the relevant roaming architecture for LBO as depicted in the figures below [36].

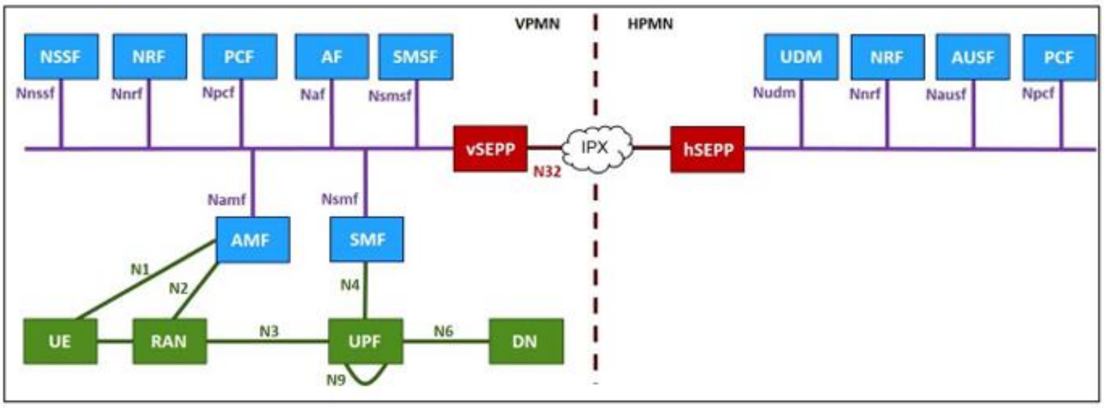

Figure 29 – 5G SA Roaming Architecture: Service Based Representation

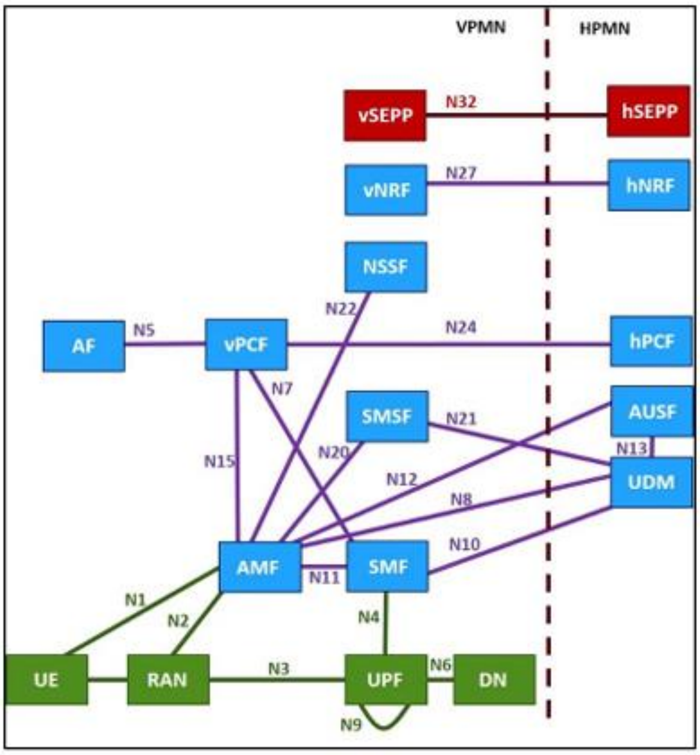

Figure 30 – 5G SA Roaming Architecture: Reference Point Representation

While the theoretical background is set, the practical implementations are still lagging due to the complexity of implementing and streamlining all the essential technological innovations. The white paper prepared by SNS IA on the CAM perspective towards 6G [18] provides a good synopsis of the technological challenges that are foreseen regarding seamless connectivity during network handovers, such as:

- Time Synchronization among network nodes (gNBs, edge sites, core functions, etc.)
- Inter-PLMN handover settings
- Scalability Issues
- Network Interconnections

ENVELOPE addresses the roaming implications through the Greek site's use case working on a prototype solution that addresses the need of a roaming framework with Local Breakout (LBO) of traffic. This is required to support seamless service migration for UEs that move across different mobile networks, and whose user data is provisioned exclusively in their original network, the HPLMN. Such a solution is targeting to combine innovative service management at the MEC level, dedicated radio handover mechanisms, and ad hoc interaction between the two core networks involved in the prototype: the HPE 5GC (as part of the HPLMN) and Open5GS (as part of the VPLMN). The proposed LBO-enabling inter-core network integration mechanism is not described in this document due to existing confidentiality requirements on its actual implementation within the HPE 5GC. Nonetheless, further details on the deployed solution and its inner mechanisms may be provided at a later stage of the project's lifetime, together with the experimental results of use case activities at the Greek testbed site.

5 Conclusions

Deliverable D2.2 has provided a comprehensive overview of the architectural framework for the ENVELOPE 6G-IA SNS Project. Through detailed analysis and systematic methodology, the requirements necessary to support the dynamic and reconfigurable nature of the ENVELOPE platform have been identified and formalized. The document has outlined the principles and motivations behind the project, emphasizing the importance of openness and innovation in the Beyond 5G (B5G) landscape.

Key architectural enablers and specifications have been explored, including Experimentation as a Service, Management and Orchestration, Edge Computing, Multi-connectivity and Predictive QoS. These components are crucial for the successful deployment and operation of advanced vertical services that leverage the capabilities of the ENVELOPE Platform.

The deliverable has also highlighted the importance of collaboration among consortium partners and stakeholders to ensure the completeness and accuracy of the requirements. By mapping these requirements to specific use cases, we have ensured their relevance and applicability to real-world scenarios.

In conclusion, the architectural framework presented in this deliverable sets the foundation for the design and implementation phases of the ENVELOPE project. It provides a clear roadmap for achieving the project's objectives and addressing the key demands of future network environments. Moving forward, the insights and recommendations from this deliverable will guide the development of innovative solutions that enhance the performance, reliability, and flexibility of B5G systems

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